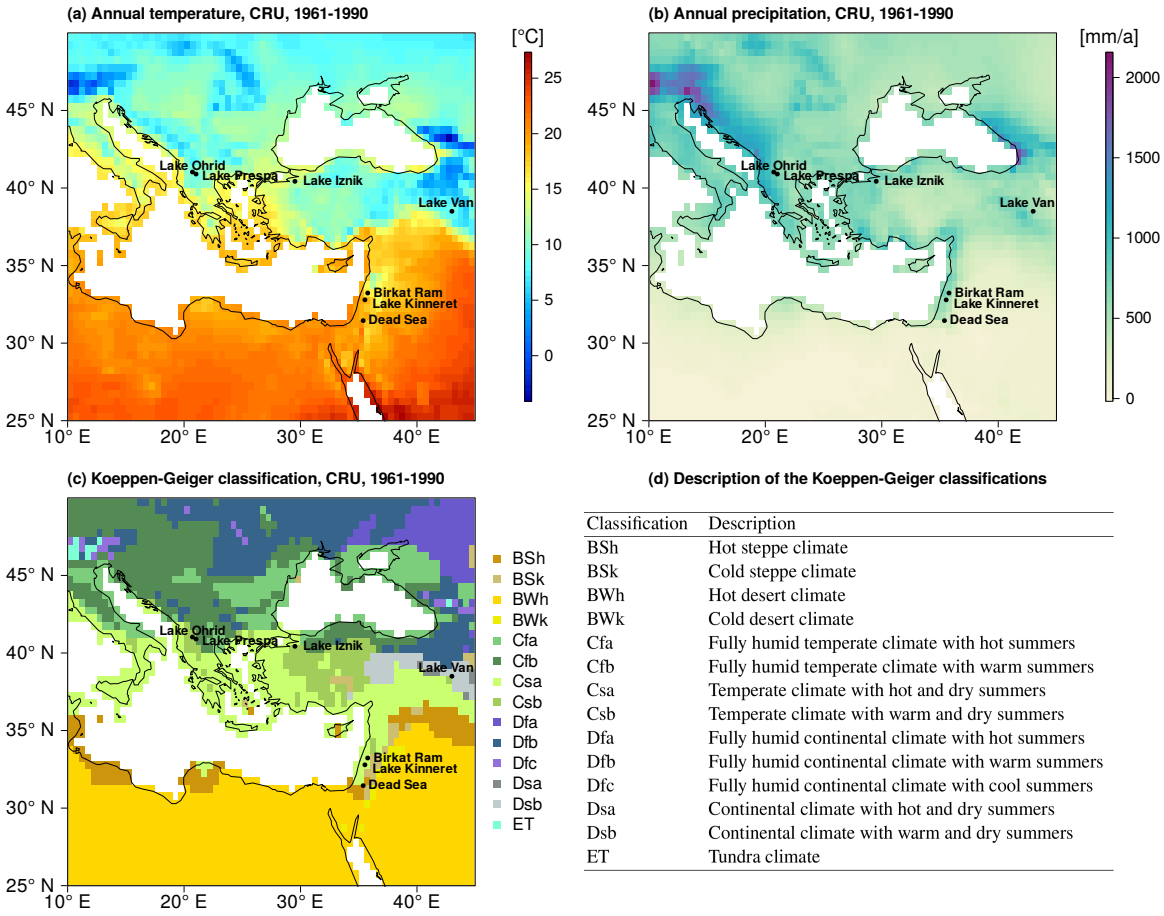
3.3 Local climate reconstructions

In Sect. 3.2 we explain how we compute the final local reconstructions and apply this procedure to the example of Lake Prespa. Here, we will present and discuss other local reconstructions. The respective results of the Bayesian age-depth models performed with Bacon are shown in Appendix A. First, some general information about the regional climate are given. Fig. 3.17 (a) and (b) show annual temperatures and precipitation for parts of southeastern Europe and the Near East. These climate values are from the CRU dataset with the reference period 1961 – 1990 AD. As we have already seen with the Lake Prespa example, we use estimated Koeppen-Geiger climate classifications for additional assessment of the reconstructed climate. The classifications based on our reference period are shown in Fig. 3.17 (c). Using the table in Fig. 3.17 (d), we can see that the sites presented have a wide



Classification	Description
BSh	Hot steppe climate
BSk	Cold steppe climate
BWh	Hot desert climate
BWk	Cold desert climate
Cfa	Fully humid temperate climate with hot summers
Cfb	Fully humid temperate climate with warm summers
Csa	Temperate climate with hot and dry summers
Csb	Temperate climate with warm and dry summers
Dfa	Fully humid continental climate with hot summers
Dfb	Fully humid continental climate with warm summers
Dfc	Fully humid continental climate with cool summers
Dsa	Continental climate with hot and dry summers
Dsb	Continental climate with warm and dry summers
ET	Tundra climate

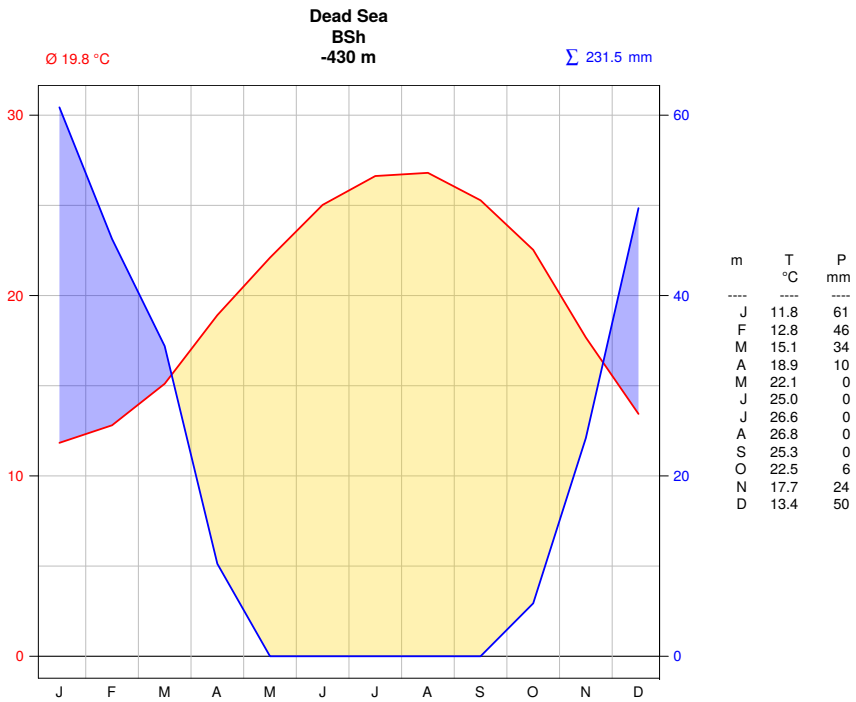
Figure 3.17: Annual temperature in (a) and precipitation in (b) based on CRU 4.01 data (Harris and Jones, 2017) and the 1961 – 1990 AD reference period. The black dots mark the locations for which the reconstructed paleoclimate is discussed in detail in this work. Panel (c) shows the corresponding Koeppen-Geiger climate classifications and the table in (d) describes them based on Chen and Chen, 2013.

variety of climate classes. This ranges from the steppe climate in the Levant to the continental climate in Eastern Anatolia to the Mediterranean and maritime climate in the Balkans. The corresponding explanations can be found in the sections listed below. These are arranged counterclockwise compared to the locations shown in Fig. 3.17.

The descriptions of the following local reconstructions always follow a similar structure. This should make it easier for the reader if they are only interested in specific reconstruction sites. In addition, the results of each site are summarized at the end of the corresponding discussion. For this reason, no additional summary is provided at the end of this Sect. 3.3.

3.3.1 Dead Sea

The Dead Sea is located at the deepest point of the continents, about 430 m below mean sea level (bmsl). This region lies on the Dead Sea Transform, a tectonic boundary between the Sinai Plate and the Arabian Plate (Miebach et al., 2019). It is a terminal lake and is fed primarily by the Jordan River, which flows year-round, but also by groundwater and several streams. The Dead Sea basin has been continuously subsiding since its formation in the early Miocene and has served as an important



m	T °C	P mm
J	11.8	61
F	12.8	46
M	15.1	34
A	18.9	10
M	22.1	0
J	25.0	0
J	26.6	0
A	26.8	0
S	25.3	0
O	22.5	6
N	17.7	24
D	13.4	50

Figure 3.18: Climate diagram of the Dead Sea region. The data are based on CRU 4.01 (Harris and Jones, 2017) and represent the grid point which is closest to the Dead Sea over the period from 1961 – 1990 AD. The red colors and numbers represent the temperatures in °C and the blue ones the precipitation in mm. The letters on the x-axis indicate the months, which are also shown on the right side of the diagram, where the climate data are listed in a table.

sediment trap. The climate diagram in Fig. 3.18 of the grid point closest to the Dead Sea shows a hot steppe climate BSh. The mean annual temperature in the Dead Sea region is approximately 20 °C, and the mean annual precipitation is around 230 mm. Summers are dry and hot, while winters have a mild and humid climate. Most of the rainfall in this region occurs between October and April. This hyper-dry climate leads to a predominance of Saharo-Arabian desert vegetation (Miebach et al., 2019). The unique orography in this region leads to a strong precipitation and temperature gradient. Therefore, the surrounding mountains are dominated by Irano-Turanian steppe and Mediterranean woodland (Miebach et al., 2019). Overall, this site is characterized not only by a transition zone of different climatic classes (cf. Fig. 3.17 (c)), but also by diverse vegetation due to the size of the Dead Sea catchment area (Litt et al., 2012).

The precursor of the Dead Sea, Lake Lisan, existed from 70 to 15 cal ka BP (Torfstein et al., 2013) or even 15 – 20 cal ka earlier (Neugebauer et al., 2016). In the following sections, we present two climate reconstructions for this region. The focus is firstly on the Dead Sea during the Holocene and secondly on the Dead Sea and Lake Lisan during the period from the Penultimate to the Last Glacial.

Holocene

Introduction In 1997 AD, a 21 m sediment core was recovered during a drilling campaign on the west coast of the Dead Sea near En Gedi (Schölzel, 2006). This contains the palynological data for the last 10,000 years, which serves as one climate proxy for the reconstruction described below. Although the top layer studied reflects the climate circa 1955 AD, we suggest that the range of the Koeppen-Geiger classification BSh for the hot steppe takes this assumption into account. We also assume that the choice of such a climate classification better reflects the catchment area of the taxa source in question than just one grid point. Therefore, this climate information is used to anchor the recent time slice in the reconstruction.

A problem arises when examining this core. It largely reflects human activity, with various cultures influencing the vegetation in the Dead Sea basin. The AP/NAP reference curve in Fig. 3.19 (a) also reveals this information. For this reason, we include another climate proxy, isotopic data of speleothems from the nearby Soreq Cave (see Fig. 3.19 (b)). In Sect. 2.1.1, the description of these data are linked to precipitation in the Eastern Mediterranean. Although the temporal space of this reference could be used for most periods of the Holocene (Bar-Matthews and Ayalon, 2007; Bar-Matthews and Ayalon, 2011), we choose to use the pattern correlation of the respective WPSs. This is due to the fact that the signal also reflects the composition of the source water of the Mediterranean Sea (Litt et al., 2012). In this case, correlation in temporal space poses some problems, but moving to the appropriate spectral space allows us to circumvent them and take advantage of the variability within these data. For all reconstructions within the Holocene and Levant, we choose the period 0 – 6.5 cal ka BP as a zone of non-negligible human impact (Schiebel, 2013). To account for this, the AP/NAP curve is used as a guide to annual precipitation only for ages prior to 6.5 cal ka BP. In

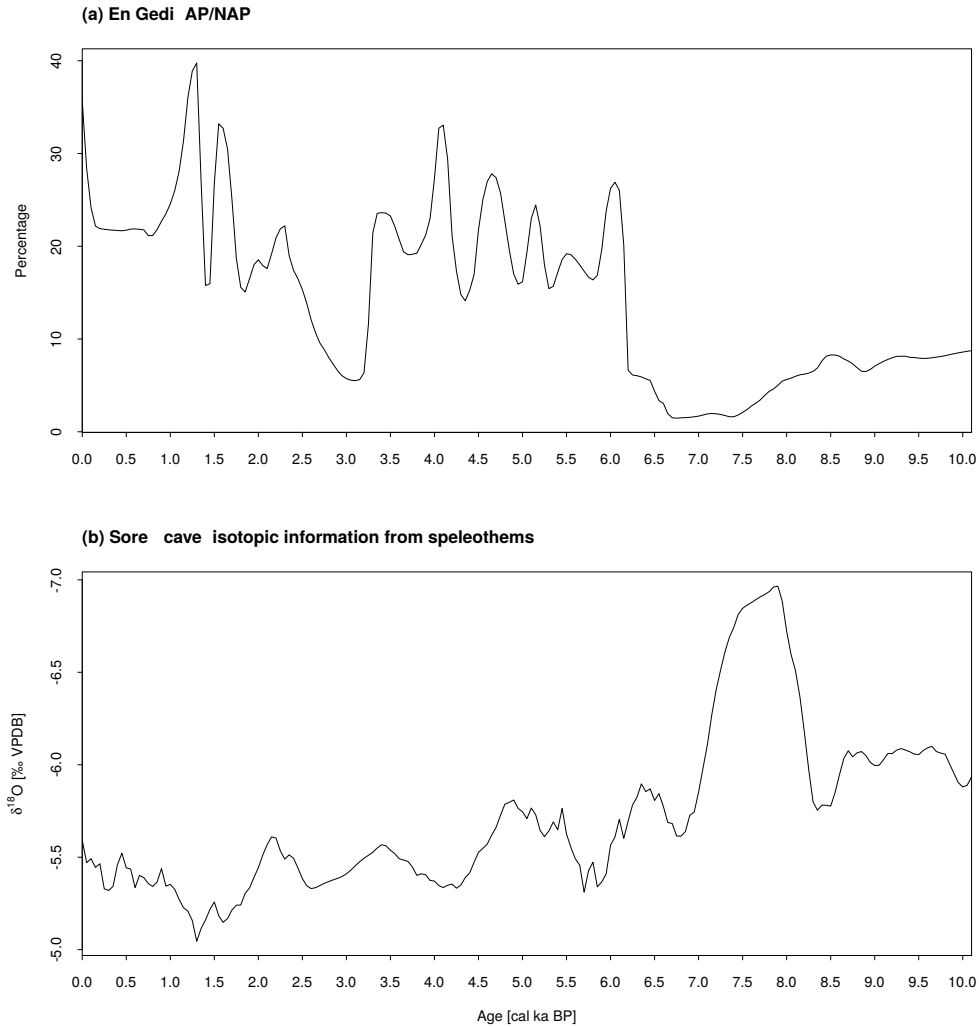


Figure 3.19: Reference curves based on (a) total terrestrial pollen sums: arboreal and non-arboreal pollen taxa from Litt et al., 2012 and (b) on isotopic information from speleothems from Soreq Cave (Bar-Matthews et al., 2003). The latter is expressed in δ^{18}O isotopic ratios in parts per thousand relative to the VPDB scale.

contrast, this climate variable is correlated with patterns of the WPS from the Soreq Cave throughout the reconstruction period.

The age-depth model data can be found in Litt et al., 2012. They are inserted into the Bayesian-based age-depth model Bacon and the results are used to convert depths to ages. These 20 radiocarbon dates indicate mostly continuous sedimentation, although they assume a hiatus at about 16 m core depth. The Bacon model is able to account for such abrupt events.

Reconstruction In the following, we describe the final reconstruction shown in Fig. 3.20. Panel (a) displays the reconstructed annual temperature T_{ANN} in °C and (b) the annual precipitation P_{ANN} in

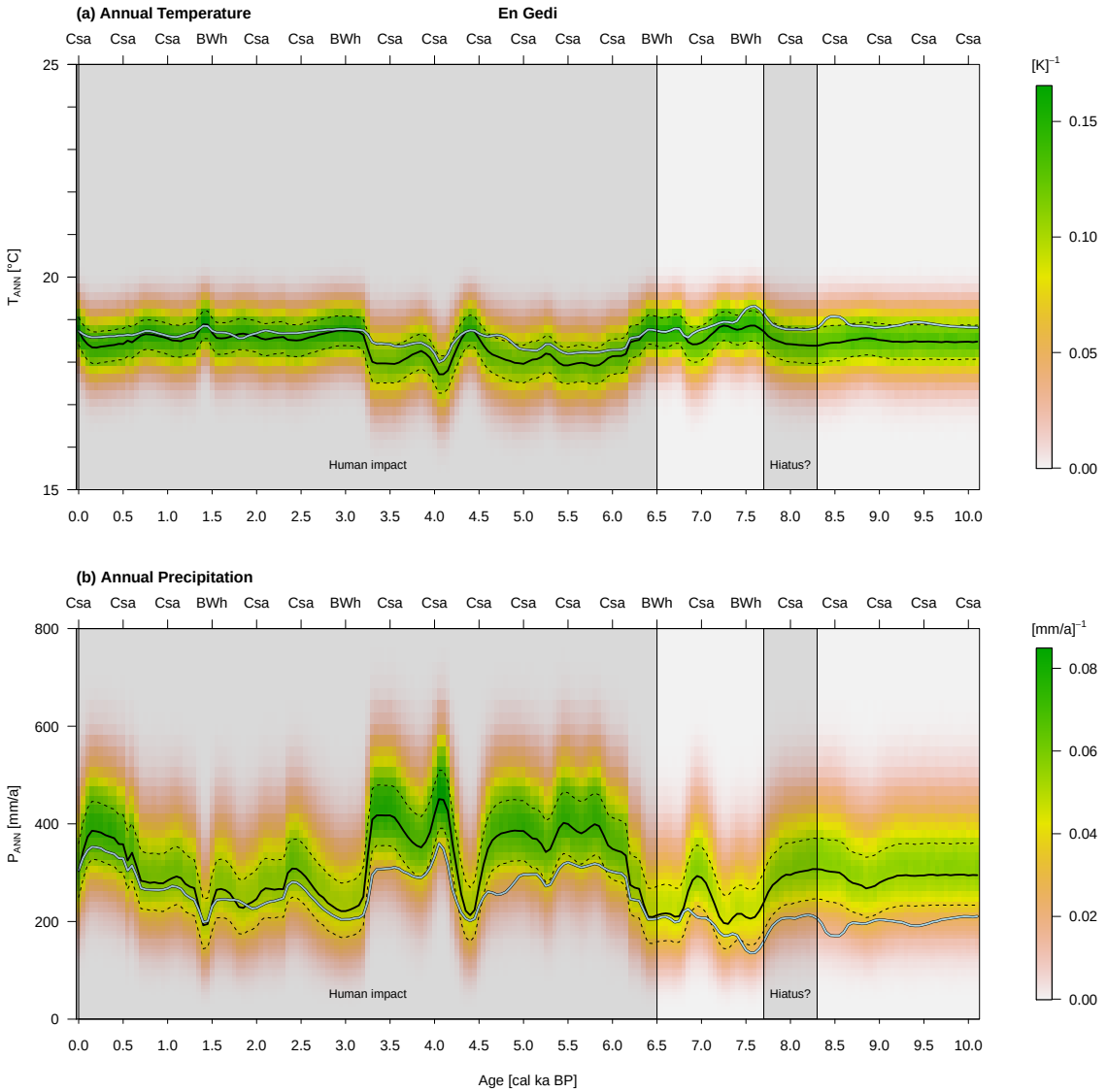


Figure 3.20: Quantitative reconstruction of the Dead Sea region based on plant and isotopic information in terms of calibrated kiloanni before 1950 AD (cal ka BP). In (a) is the reconstructed annual temperature in °C and in (b) is the annual precipitation in mm/a. The colors indicate the probability density values, the black solid lines its median, and the dashed black lines the first and third quartiles. The blue lines represent the medians of the reconstructions without additional information. The grayed out area on the left shows the zone of increased human impact and on the right the ages that contain approximately a hiatus. The axis above the graph shows the approximated Koeppen-Geiger classifications corresponding to the PDFs for the respective periods.

mm/a. At the beginning of the explanations, we focus on the solid black lines. Later, we will work out the difference between these reconstructions and others whose medians are shown as blue lines. Litt et al., 2012 divided their vegetation history into different age ranges. These are also used for this study.

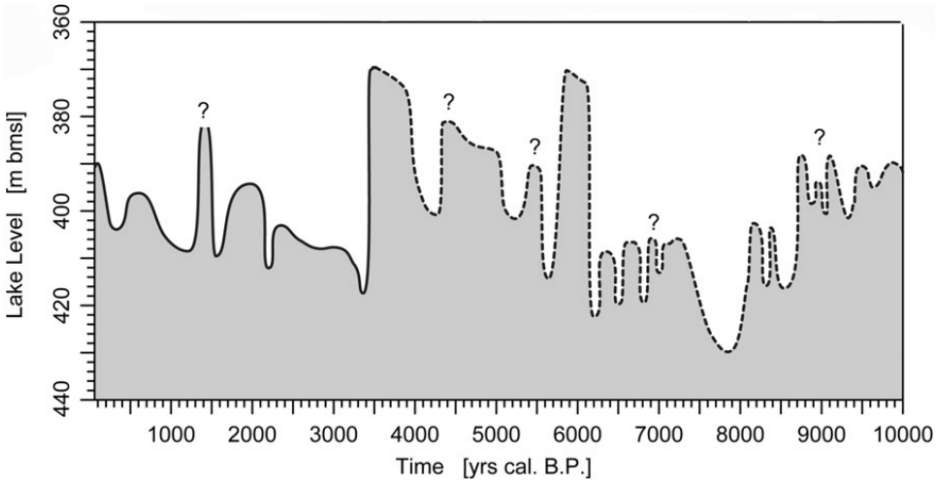


Figure 3.21: This figure is from Litt et al., 2012 and depicts the independently determined height of the Dead Sea lake level based on the work of Migowski et al., 2006.

First, some **general remarks** are made. Our reconstructed precipitation and temperature behave strongly anti-correlated with a value of about -0.95. This is due to the corresponding transfer functions, where higher temperatures are associated with lower precipitation. In the Levant, this is a typical pattern (cf. Fig. 3.17 (a) − (c)), as the hotter deserts and steppes (BWh/BSh) have on average much less precipitation than the colder Mediterranean coastal regions (Csa). When we describe a decrease in precipitation, it is always associated with an increase in annual temperature, unless otherwise noted. Although Litt et al., 2012 reconstruct winter and not annual temperature, such an anti-correlation also exists in their climate reconstruction.

The **10 – 7.6 cal ka BP** time interval spans the transition from Pre-Pottery Neolithic to Pottery Neolithic at ca. 8.4 cal ka BP (Schiebel, 2013) and Litt et al., 2012 describe it with low tree pollen values. These low percentages in Fig. 3.19 (a) correspond to the relatively low reconstructed lake levels in Fig. 3.21, including the notable decline from about 8.2 cal ka BP. The influence of humans during these ages could therefore be neglected. The authors assume a hiatus during these periods, which is shown in Fig. 3.20 with the right grayed out area. The reconstructed annual precipitation in (b) shows relatively low values around 300 mm. Together with the respective temperature around 18.5°, this section lies within the hot Mediterranean Koeppen-Geiger classification Csa. The AP/NAP reference curve in Fig. 3.19 (a) shows a very similar pattern to P_{ANN}. Bar-Matthews et al., 1999 also describe this with a decrease in precipitation. In our reconstruction, the decrease in P_{ANN} starts relatively slowly at first at the same time.

For the remainder of the Pottery Neolithic (**7.6 – 6.3 cal ka BP**), Litt et al., 2012 describe a decline in Mediterranean trees and shrubs. They conclude that the Mediterranean vegetation belt has shifted northward and human influence is relatively low due to low settlement. The reconstructed precipitation shows the lowest and the temperature the highest values of the whole reconstruction period. This is

associated with a change from the hot Mediterranean climate to the hot desert climate BWh, which underscores the conclusion of Litt et al., 2012. Both reference curves show their minimum during these periods. The AP/NAP curve increases by about 6.5 cal ka BP, while the isotope-based curve does so 1000 years earlier. This could be the cause of the change to Csa by 7 cal ka BP and the return to BWh by 6.5 cal ka BP. The quantitative climate reconstruction in Litt et al., 2012 shows a reduced Mediterranean biome prior to 6.5 cal ka BP with positive winter temperature anomalies and an expected annual precipitation value of 280 mm/a. As for the trend analysis, their reconstruction shows a negative trend in P_{ANN} for these years. Such features are also evident in our new reconstruction of En Gedi.

The next time span of **6.3 – 3.2 cal ka BP** includes the beginning of the Chalcolithic (ca. 6.5 – 5.5 cal ka BP), the Early and Middle Bronze Age (ca. 5.5 – 4.15 and ca. 4.15 – 3.5 cal ka BP), and the Late Bronze Age (ca. 3.5 – 3.15 cal ka BP) (Schiebel, 2013). Litt et al., 2012 note an increase in Mediterranean woodland, which could be due to an increased ratio of precipitation to evapotranspiration. However, human impact increases during this period. For this reason, the reconstructed P_{ANN} is no longer correlated with the AP/NAP curve. As a guideline, we are left only with the pattern correlation of this climate variable with the WPS from Soreq Cave. The reconstructed precipitation shows the highest values with a maximum around 450 mm during this period. On the other hand, the annual temperature drops to almost 17.5 °C. This change in climate is also reconstructed in Litt et al., 2012, consistent with relatively high lake levels. They found precipitation minimums at about 5.6 and 4.3 cal ka BP. In comparison, our reconstruction shows minima of P_{ANN} at about 5.4, 4.3, and 3.8 cal ka BP. Since these declines sometimes last several hundred years, the archaeological transitions described above could be related to these climate fluctuations. Bar-Matthews and Ayalon, 2007 also connect the corresponding variations in Fig. 3.22 (b) to cultural changes in the Eastern Mediterranean and to the periodic behavior of Bond events. This can be seen in the corresponding WPS in Fig. 3.22 (a), which shows a significant power branch with a period of about 1.4 ka. Comparing this with the WPS of P_{ANN} in Fig. 3.22 (c), two branches with significant power of about 1 ka and 2 ka periods stand out at the beginning of this time interval. Later, both merge to a period of about 1.5 cal ka BP. This spectral pattern is also evident in the WPS of IRD in the North Atlantic from Bond et al., 2002 in Fig. 2.7 (d), underscoring that our reconstruction has similar variations to some Bond events.

The next period of **3.2 – 2.3 cal ka BP** begins with the transition from the Late Bronze Age to the Iron Age. Litt et al., 2012 notes a dramatic decline in the Mediterranean biome, reconstructed precipitation, and lake level, that cannot be explained by human influences. Our reconstruction shows a similar pattern and a change to a hot desert climate. This climate change could be associated with the so-called 3.2 ka drought event and with the Late Bronze Age collapse (Schiebel and Litt, 2018). The corresponding WPS in Fig. 3.22 (c) shows the highest power with a period around 1.5 ka at the beginning of this period. At the end of the Iron Age (ca. 2.6 cal ka BP), a moderate increase in oak and olive pollen is accompanied by an increase in precipitation and the transition to a hot Mediterranean

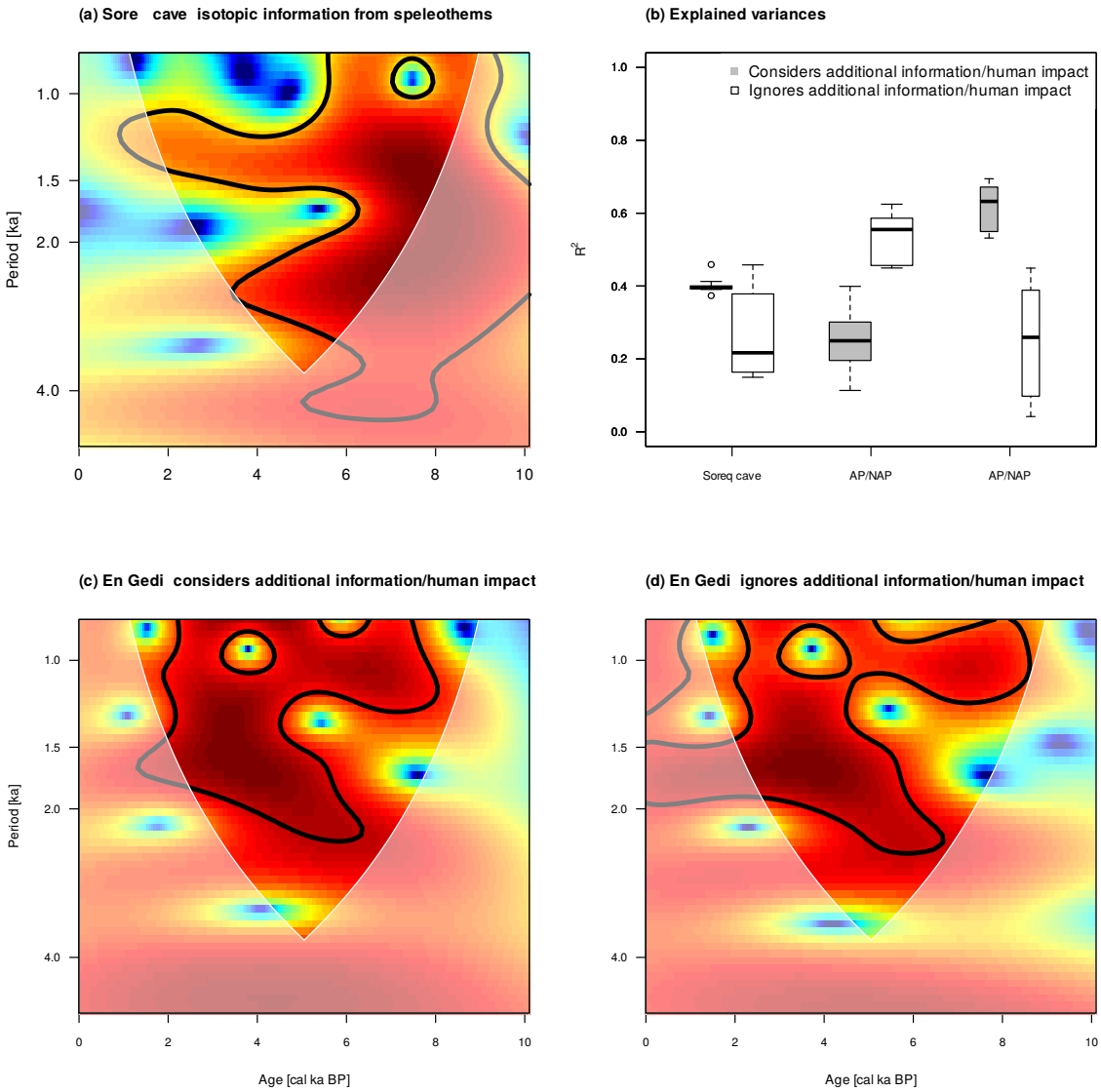


Figure 3.22: Wavelet power spectrum of (a) isotopic information from speleothems from Soreq Cave (Bar-Matthews et al., 2003), (c) the reconstructed annual precipitation with additional information, and (d) the reconstructed annual precipitation without additional information. The gray boxplots in (b) show the results with the additional information from speleothems. From left to right are plotted, first, the explained variances between the WPSs of the medians of P_{ANN} and the WPSs of the Soreq cave, and second, the medians of P_{ANN} and the entire AP/NAP curve. The thinner boxplot indicate R^2 between the medians of P_{ANN} and the section of AP/NAP where human influence is lower. The white boxplots show the same calculations, but without the additional information from the speleothems. The basis for these calculations are the selected samples of the posterior reconstruction.

climate.

In the years **after 2.3 cal ka BP**, human influence continues to increase. Orland et al., 2009, using

isotopic information from Soreq Cave, describe a decrease in precipitation during 1.9 – 1.3 cal ka BP that is also reflected in the level of the Dead Sea. They assume that this climate change weakened the economic system of the Roman and Byzantine Empires, which contributed to the decline of their rule in the Levant. This decrease in precipitation is also evident in the reconstructions of Litt et al., 2012 and in this study. At about 1.3 cal ka BP, the corresponding Koeppen-Geiger climate classification changes to BWh. Thereafter, P_{ANN} increases again, and both reconstructed climate variables approach the recent climate measurements for the youngest time interval.

Sensitive study The final reconstruction contains the pollen information from the core, including the AP/NAP, the isotopic data from Soreq cave, the recent climate measurements, and the human impact. In contrast, the reconstruction represented by the blue line in Fig. 3.20 does not take into account the additional information from the speleothems and the human influence on the vegetation.

In Fig. 3.22 (b) we can see the different explained variances. The results from the Soreq cave in the case of additional information from this site are higher (about 0.4) than the results without this isotopic information (between 0.2 and 0.4). Comparing the medians of P_{ANN} of the selected posterior

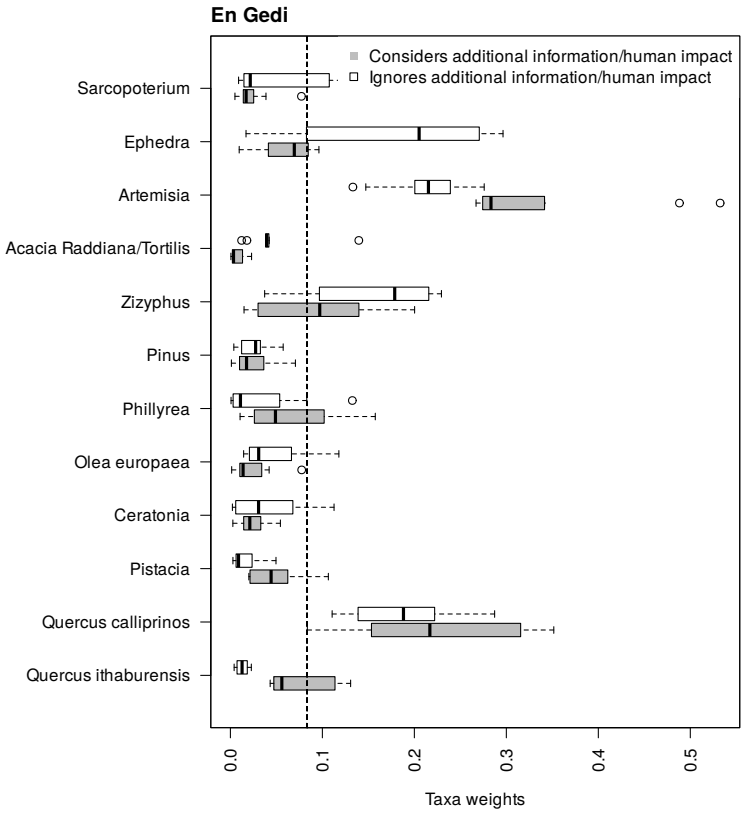


Figure 3.23: Posterior taxa weights of the reconstructions with and without additional information. The dashed line indicates the prior weights for each taxon.

samples with the whole AP/NAP curve, the corresponding R^2 is lower (from 0.15 to 0.4) than for the reconstructions with the whole AP/NAP information (about 0.55). The thinner boxplots reflect the explained variances of AP/NAP where human influence is lower. In contrast to the corresponding calculations for the entire region, R^2 is higher (about 0.65) when human influence is taken into account compared to the thin white boxplot (from 0.05 to 0.45).

This shows that the algorithm is able to properly use the information from the caves and the human influences to find a Pareto optimum between all these variables. In Fig. 3.22 (c) and (d), these differences are most pronounced for ages older than about 6 cal ka BP and thus fall within the range of lower human influence. There, the power is higher when additional information/human influences are considered, with a local maximum around 7 cal ka BP and a period of about 1000 years. Comparing this with the annual precipitation in Fig. 3.20 (b), it reflects the local maximum around this time, which is less pronounced in the blue line. As mentioned earlier, this could be due to the relatively large variation within the isotopic information. However, two taxa indicate a local maximum during these ages. Quercus ithaburensis and Pistacia show a slight increase in the pollen spectrum, which is detected by the algorithm and leads to a relatively higher weighting (see Fig. 3.23). Litt et al., 2012 associate these plants with Mediterranean vegetation, resulting in a wetter and cooler climate during this period. In addition, the reconstruction that does not take into account the information from Soreq Cave and human influences assigns a higher weight to the taxon Ephedra, resulting in the largest change in the weighting of the taxa. Moreover, the explained variance of the Ephedra pollen curve compared to AP/NAP is the highest of all plant information used, with a correlation of about -0.75. This tuning results in an overall lower R^2 of the final reconstruction with respect to the pollen reference curve. Since Ephedra belongs to the semi-arid steppe vegetation corresponding to a relatively dry and hot climate (Litt et al., 2012), the blue curve shows, on average, a reconstruction with less precipitation and higher temperatures. The maximum differences for P_{ANN} are about 100 mm and for T_{ANN} ca. 0.5°C.

Summary and conclusion The quantitative climate reconstruction of the Dead Sea region during the Holocene includes not only paleobotanical data, but also isotopic information from speleothems from nearby Soreq Cave. In addition, this section outlines a way to deal with the non-negligible impacts of humans on vegetation. Although the final reconstruction is based only on spectral information from speleothems for ages younger than 6.5 cal ka BP, the results of P_{ANN} are comparable to the Dead Sea lake level (cf. Fig. 3.21). However, it should be mentioned here that the fluctuations of the lake level are due to the freshwater balance evaporation - precipitation in the catchment area of the lake. Therefore, it may be beneficial in future work to additionally link plant proxies to evaporation. Furthermore, our reconstruction results are in agreement with those of Litt et al., 2012. These describe the MH (6.5 – 3.5 cal ka BP) as a relatively wet period with maximum values more consistent with our final reconstruction than with the lower precipitation that results when information from speleothems

and human influences are ignored. Moreover, the algorithm is able to find climate changes that can be associated with Bond events and known archaeological and cultural changes in the Levant (Schiebel, 2013; Bar-Matthews and Ayalon, 2011; Orland et al., 2009).

For these reasons, we conclude that the use of the above information is beneficial for such reconstructions. They can be easily extended by other proxy information such as lake levels or climate constraints from other quantitative reconstructions. In these cases, the first information could be used as another reference curve and the second as the data basis for additional climate anchor points. The latter option is described in Sect. 3.3.5 and a reconstruction without any climatic anchors is discussed in the following section.

Late Penultimate to Last Glacial

Introduction In this section, we aim to reconstruct the climate of the Dead Sea region over a period of about 14.5 – 147 cal ka BP. Therefore, we cannot use information from recent climate measurements for the youngest time slice of this reconstruction period. Thus, this reconstruction tests whether the algorithm is capable of generating reasonable climate values without such information.

The reconstruction is based on pollen data presented in Chen and Litt, 2018 and Miebach et al., 2019. They investigate the deepest borehole, 5017-1-A, with a total length of 455.34 m, drilled in 2010/2011 in the northern Dead Sea basin. Unlike the previous AP/NAP references, the corresponding curve of this core is difficult to correlate with any of the reconstructed climate variables. Miebach et al., 2019 describe the amount of arboreal pollen as an indicator of available moisture, which depends mainly on precipitation and evapotranspiration. In particular, when studying a period that includes glacials and interglacials in this region, information on evapotranspiration is needed to use the AP/NAP curve as a reference. However, since we only consider T_{ANN} and P_{ANN}, we will not use this reference curve for the reconstruction. Instead, we include isotopic information not only from Soreq Cave but also from the Mediterranean Sea, which is shown in Fig. 3.24. The latter are grouped together in the so-called Medstack, which comprises several locations (Colleoni et al., 2012) and can be linked to the sea surface temperature of the Mediterranean Sea (Wagner et al., 2019). Therefore, we use this proxy as a guideline for annual temperature in the Dead Sea region and the WPS of the Soreq Cave for the reconstructed annual precipitation.

The data for the Bayesian-based age-depth model can also be found in the studies of Chen and Litt, 2018 and Miebach et al., 2019. For most dates younger than about 44 cal ka BP, the uncalibrated radiocarbon ages are used and fitted to the Bacon model. The corresponding errors are relatively small, resulting in a much sharper age-depth transformation than for ages older than 40 cal ka BP. This section mainly uses the calendar ages of the uranium-thorium data from Dead Sea Core 5017-1-A. For ages older than about 120 cal ka BP, Chen and Litt, 2018 use the anchor ages of LR04 to correctly determine the transition from MIS 6 to MIS 5.

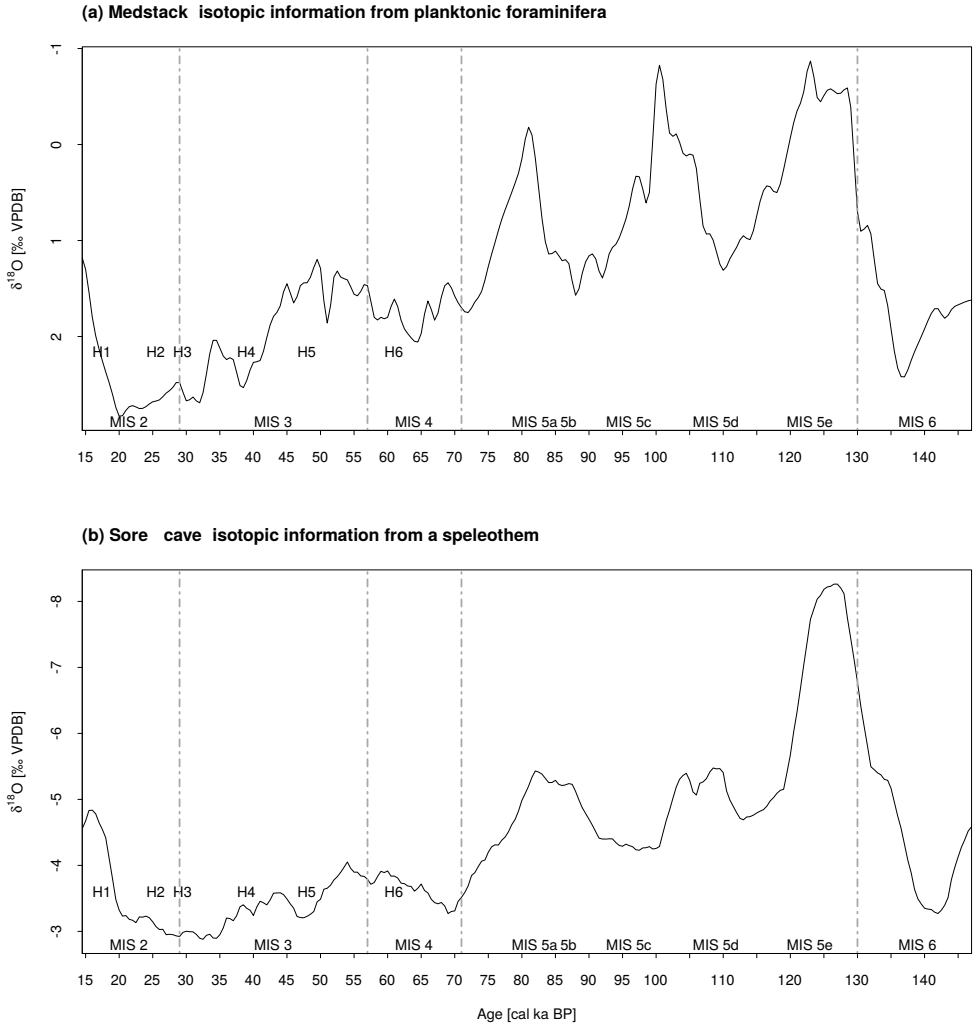


Figure 3.24: Reference curves based on (a) isotopic information from planktonic foraminifera from stacked marine cores of the Mediterranean Sea (Colleoni et al., 2012) and (b) on isotopic information from a speleothem from Soreq Cave (Grant et al., 2012). Both proxies are expressed in δ^{18}O isotopic ratios in parts per thousand relative to the VPDB scale. H 1 – 6 describe the relative timing of Heinrich stadials according to Hemming, 2004 and MIS 2 – 6 the marine isotopic stages following Lisiecki and Raymo, 2005. The vertical gray lines indicate the transitions of the MISs.

Reconstruction In the following, we describe the final reconstruction shown in Fig. 3.25. Panel (a) displays the reconstructed annual temperature T_{ANN} in °C and (b) the annual precipitation P_{ANN} in mm/a. At the beginning of the explanations, we focus on the solid black lines. Later, we will work out the difference between these reconstructions and others whose medians are shown as blue lines. Chen and Litt, 2018 and Miebach et al., 2019 have divided their explanations into different age ranges. These are also used for this study.

For this reason, we start with the calibrated years between **147.3 – 130.9 cal ka BP**. These include

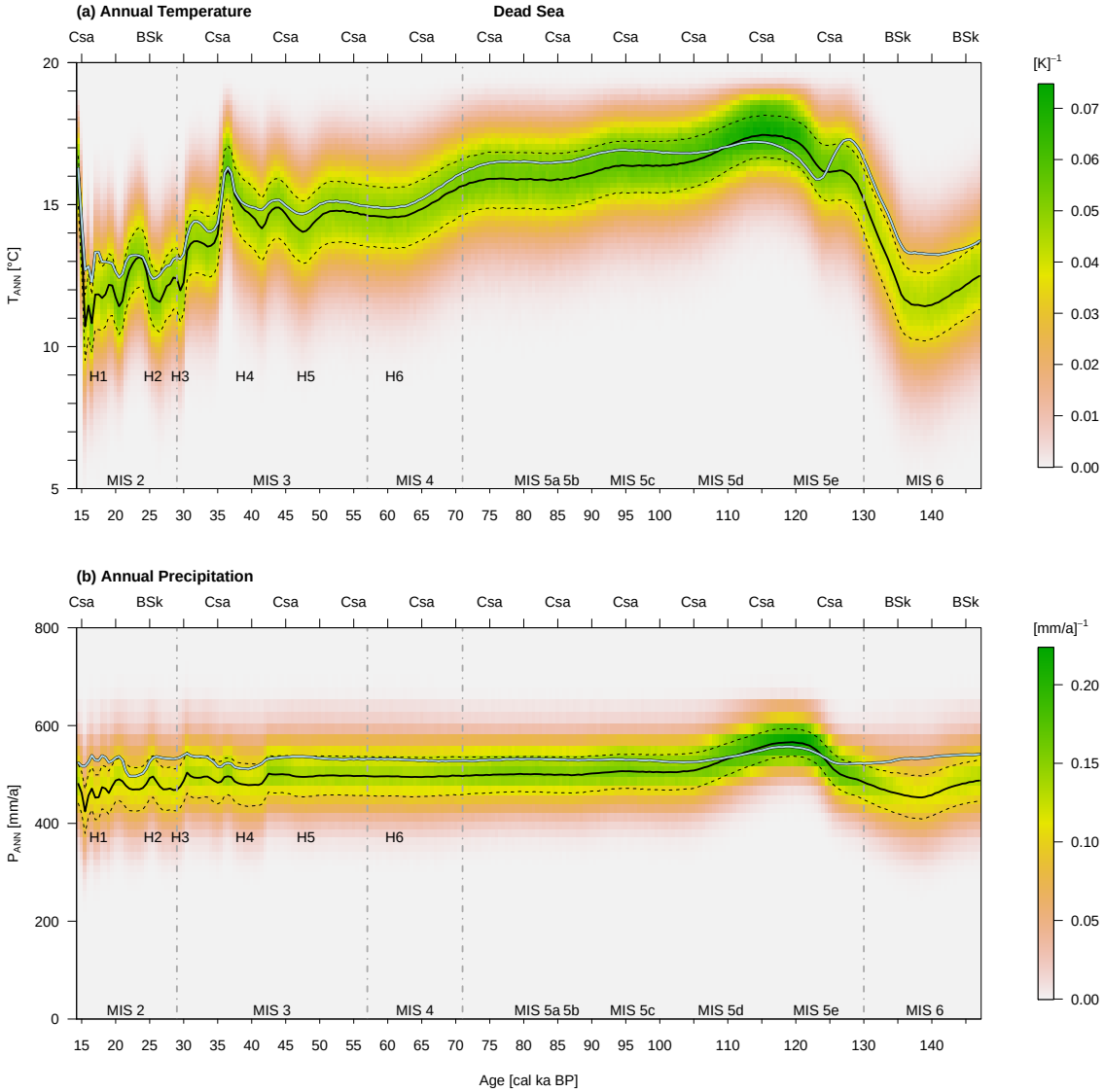


Figure 3.25: As Fig. 3.20, but for the Dead Sea. H 1 – 6 describe the relative timing of Heinrich stadials according to Hemming, 2004 and MIS 2 – 6 the Marine Isotope Stages following Lisiecki and Raymo, 2005. The vertical gray dashed dotted lines indicate the transitions of the MISs.

the late Penultimate Glacial and late MIS 6. Chen and Litt, 2018 describe that the vegetation is dominated by wormwood trees and that the proportion of trees and shrubs is relatively moderate. The reconstructed T_{ANN} averages 12.25 °C and P_{ANN} 470 mm. During 134 – 140 cal ka BP, we find the respective minimums at about 11.5 °C and 450 mm. Inserting these values into the transfer function $\mathbb{P}(\boldsymbol{P_s}|\boldsymbol{C},\boldsymbol{\psi})$ of Quercus ithaburensis and Quercus calliprinos, we obtain probabilities of approximately 45 % and 50 %, respectively. This is consistent with research by Chen and Litt, 2018 that reduced

temperature still supports survival of temperate trees. They also conclude that this reduction leads to low evaporation rates, resulting in a positive freshwater balance in the lake's catchment. Progressive aridification during the Penultimate Glacial in the Dead Sea region results in the Koeppen-Geiger approximation BSk. For comparison, this cold steppe climate is found today in some parts of Central Anatolia (cf. Fig. 3.17 (c) and (d)), where steppe vegetation predominates (Kurt et al., 2006). Looking at the isotopic data from Medstack and the Soreq cave, a very similar pattern emerges during MIS 6 with local maxima. Nevertheless, a displacement of these by about 5 ka is present. This shift is also found in the climate reconstruction, where annual temperature increases at about 135 cal ka BP and annual precipitation ca. 5 ka later.

At the beginning of the **130.9 – 115.5 cal ka BP** period, which includes the Last Interglacial optimum and corresponds to MIS 5e, frost-sensitive pistachio pollen increases (Chen and Litt, 2018). This should be accompanied by an increase in temperature, which confirms our reconstruction model. T_{ANN} is about 16 °C and P_{ANN} is about 500 mm. After ca. 125 cal ka BP, there is a further increase in both climate variables. Here Chen and Litt, 2018 identifies the onset of the LIG optimum, which duration is about 124.2 – 115.5 cal ka BP. This is coupled with an increase in grasses, trees and shrubs in the mountains. The reconstructed climate values have their maximum in this period with values around 18 °C and 570 mm. This rise in precipitation contrasts with the course of the reconstructed lake level (Torfstein et al., 2015), which shows a decline during MIS 5e. Chen and Litt, 2018 explain this with the increased vegetation mentioned above, leading to enhanced evapotranspiration in the area of the main headwaters of the Dead Sea. This could lead to a decrease in freshwater supply and thus to a drop in the lake level, even though precipitation has increased. Thus, not only the correlation with AP/NAP proves difficult, but also the comparison with the reconstructed lake level. After about 120 cal ka BP, there is a variation in the pollen spectrum with relatively constant pistachio pollen but a decline in Mediterranean sclerophylls. This change does not affect our climate reconstruction because their mean values remain comparatively stable. With respect to MIS 6, the Koeppen-Geiger classification of the Dead Sea region changes to a hot Mediterranean climate Csa throughout the MIS 5 period. This classification occurs most frequently in the climate reconstruction of En Gedi during the Holocene. Although the reconstructions do not include information on interannual climate, the Koeppen-Geiger approximation suggests mild/wet winters and hot/dry summers. This is consistent with the statements of Chen and Litt, 2018, which indicate the same climate. Furthermore, the reconstructed temperatures show similar values to those of the Holocene. In contrast, precipitation appears to be somewhat higher. This is in agreement with the results of Bar-Matthews et al., 2017, which indicate wetter conditions during the LIG in the southern Levant than today. Both isotope-based reference curves show their extreme values during the same period with an increase after 120 cal ka BP. In doing so, Medstack does not rise as steeply as the Soreq cave, resulting in a shift in extrema during the remainder of MIS 5.

The early Last Glacial within **115.5 – 89.1 cal ka BP** includes a decline in pistachio and an almost complete absence of sclerophyllous pollen (Chen and Litt, 2018). From this, they infer a cooling

trend that could be associated with the MIS 5d stadial. During the period 105-99.6 cal ka BP, there is a slight rebound of deciduous oaks. Chen and Litt, 2018 interpret this period with the interstadial MIS 5c. Both climate variations are seen for both climate variables in our reconstruction, with the corresponding amplitudes in P_{ANN} being relatively small. Since 99.6 ka, the steady decline in pistachio populations has been accompanied by a decline in both climate variables. This could be indicative of stadial MIS 5b. As mentioned earlier, the alternating MIS 5 stadials and interstadials are shifted in both isotope-based references. Nevertheless, the shift of MIS 5d with respect to the climate reconstructions is much smaller and in the opposite direction than indicated in the guidelines. In contrast, the interstadial MIS 5c fits these marine isotope data quite well, although they are not known to the algorithm in this case. This behavior emphasizes the restriction of the explained variances to not too high values, which allows the reconstruction method to find a compromise between all included proxy information.

Miebach et al., 2019 describe the age **88 – 62.6 cal ka BP** with rapid paleoenvironmental changes, with a dominance of herbs and dwarf shrubs and some moderate Mediterranean woodland elements. In addition, the pollen spectrum could contain a Dansgaard-Oeschger signature. In our reconstruction, both climate variables show stable behavior during the end of MIS 5b and during MIS 5a. It should be noted again that the age-depth transformation leads to a relatively blurry picture due to the high uncertainties in these periods. Therefore, the information about such rapid climate changes may not be resolved. After the transition to MIS 4 at about 72 cal ka BP, there is a drop from 16 °C to 14.5 °C in T_{ANN}. The local minimum at about 63 cal ka BP coincides with the Heinrich stadial 6 (Hemming, 2004). The corresponding amplitudes in P_{ANN} continue to decrease. The reference curves show a relatively correlated pattern, although the timing of H 6 in Fig. 3.24 (a) is more shifted than in Fig. 3.24 (b).

The years **62.6 – 34.7 cal ka BP** contain the late MIS 4 and the early/mid MIS 3. Miebach et al., 2019 characterize this period with the strongest variation in vegetation between MIS 5 and MIS 1 with an increase in AP and the highest mean percentages of Quercus ithaburensis. Relatively high fluctuations can also be observed for T_{ANN}. The variance of P_{ANN} increases again after about 42 cal ka BP, indicated by lower values of the probability density within the uncertainty ranges. Prior to this, T_{ANN} drops by about 1 °C at about 47 cal ka BP, which is consistent with H 5. This deviation is not as clearly visible in the corresponding reference curve (see. Fig. 3.24 (a)) as in the Soreq cave. This is followed by a temperature increase with a subsequent analogue decrease at about 42 cal ka BP. In contrast to the previous decline, this can be seen in the Medstack curve, but not so clearly in the patterns of the speleothem. Furthermore, P_{ANN} also shows a decrease. These climate fluctuations could be related to H 4, which occurs around 40 cal ka BP. The corresponding WPS of P_{ANN} in Fig. 3.26 (c) shows a relatively high non-significant power after about 45 cal ka BP with periods between 1 and 3 ka. These variations could include some Dansgaard-Oeschger (9 – 11) events (cf. Fig. 2.3). From now on, the age-depth model has a much lower variance, which results in a sharper age-depth

transformation. In this way, the rapid climate fluctuations can be resolved. After these fluctuations, both variables increase again until about 35 cal ka BP.

During late MIS 3 (**34.7 – 30.6 cal ka BP**), Miebach et al., 2019 describe a complete absence of thermophilous Olea europaea and thus a reduced temperature. There are also large proportions of steppe vegetation accompanied by Mediterranean forest components. To test these observations, they model a biome shift with a decrease of $-3\,^{\circ}$C and $-15\,\%\,P_{ANN}$. These values are taken from Stockhecke et al., 2016 for MIS 3. The result is a well-mixed equilibrium between the three biomes included, the Mediterranean woodland, Irano-Turanian steppe and Saharo-Arabian desert, with a higher probability for the second biome compared to today for the Dead Sea region. Although their temperature variations are based on the mean summer and winter temperatures, we compare them to the relative differences we have reconstructed. Our reconstruction shows a mean annual temperature of about $14.5\,^{\circ}$C and P_{ANN} of 490 mm for MIS 3. The difference between these values and those of the reconstructed LIG optimum agrees quite well with those of Miebach et al., 2019. Both reference curves also show an increase for this period, accompanied by a decrease to $12\,^{\circ}$C and 470 mm. This decline marks the transition to MIS 2, along with H 3, which is consistent with the climate variations we simulated.

The period **30.6 – 15.4 cal ka BP** includes a high concentration of Artemisia and a complete absence of Olea and frost-sensitive Pistacia pollen during MIS 2 (Miebach et al., 2019). They conclude that this is the coolest period after the Penultimate Glacial with a sparse vegetation cover. Again, they model the distribution of biomes with a decrease of $-6\,^{\circ}$C and $-30\,\%\,P_{ANN}/+100\,\%\,P_{ANN}$. The results are an under-/overestimation of the Mediterranean biome. In our reconstruction, we have a mean T_{ANN} of about $12\,^{\circ}$C and P_{ANN} of 470 mm for MIS 2. This leads to differences from the LIG optimum of $-6\,^{\circ}$C and -20 %mm respectively. The reconstructed climate is comparable to that of the Penultimate Glacial. This is also indicated by the change from the hot Mediterranean climate classification Csa to the cold steppe BSk. The minimum values for both variables are about $11\,^{\circ}$C and 425 mm and are located shortly after the H 1 event. The reconstructed lake level of Lake Lisan shows the highest values during MIS 2 compared to MIS 3 and 4 (Torfstein et al., 2013). This is in contrast to the reconstructed precipitation, which shows the lowest values during MIS 2. One explanation for this behaviour could be the changed evaporation and evapotranspiration rate for this period in this region. During the LGM, Schiebel, 2013 describe a woodland clearance in the northern mountains of Israel that could reduce plant transpiration and promote a positive freshwater balance in Lake Lisan (Miebach et al., 2019). In Miebach et al., 2017, vegetation during the late Last Glacial is described for the Sea of Galilee. In their palynological study they find a similar vegetation pattern with a dominance of steppe. Due to the lower temperatures and higher relative humidity, Miebach et al., 2019 conclude that evaporation decreases, resulting in higher effective humidity. Our reconstruction, based on several proxies, underlines these considerations. Corresponding features can be seen in the isotope curves. However, the decline of the two curves starts later than in the reconstruction. Furthermore, the WPS

of P_{ANN} shows significant power for the late MIS 2 with a local maximum of a period around 1.5 ka, although the corresponding reference WPS of the Soreq cave does not contain these features. This could indicate the last Dansgaard-Oeschger event, which can also be recognized in the pollen spectrum. At the beginning of MIS 2, H 2 is accompanied by a local minimum in T_{ANN} and a local maximum in P_{ANN}. In the corresponding WPSs we can also observe a relatively high power (not significant) with a period between 1 and 2 ka. This suggests that the pollen data also contain the information from DO 3

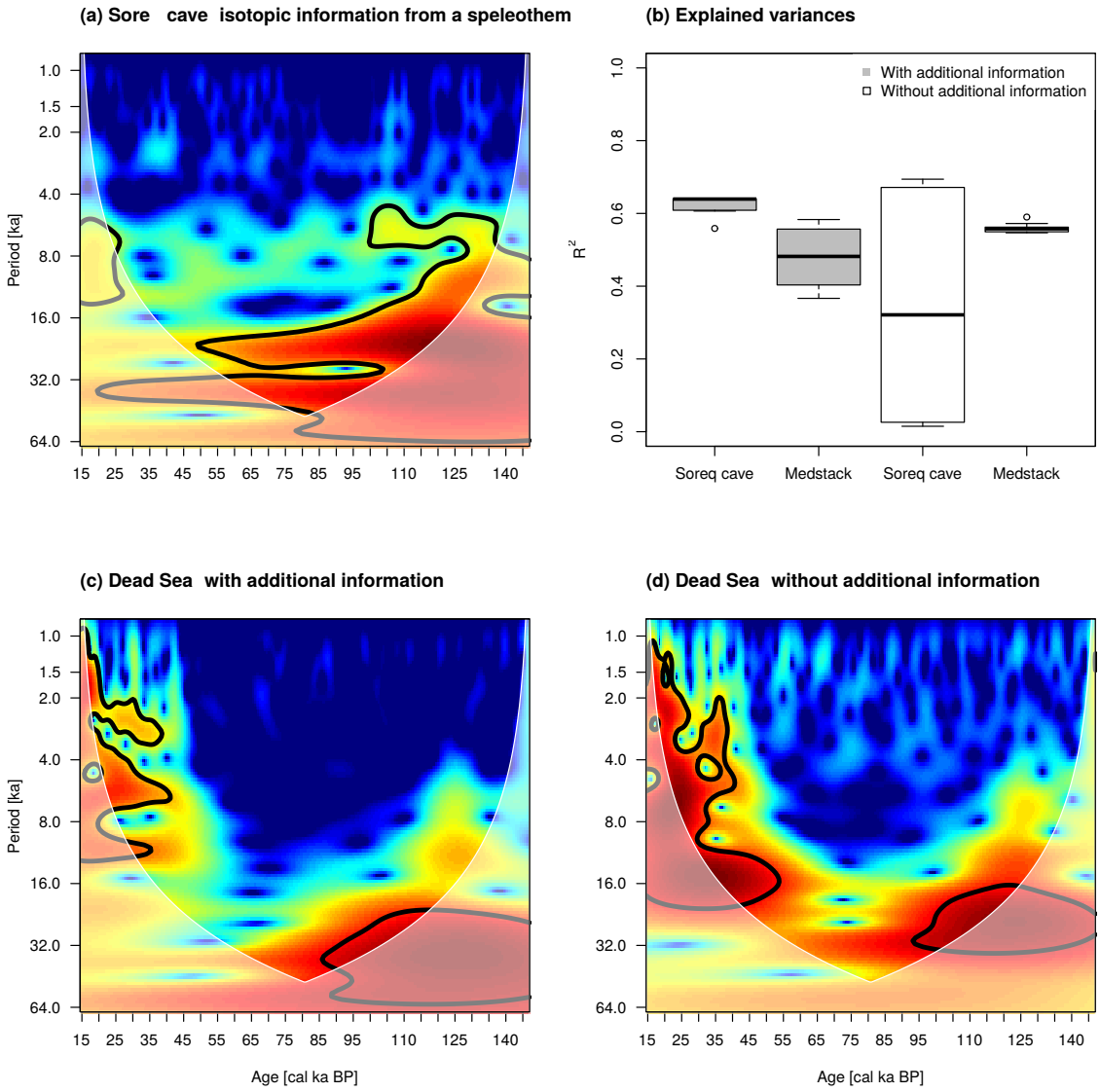


Figure 3.26: As Fig. 3.22, but for the Dead Sea. In contrast, panel (b) does not contain information on AP/NAP. Instead, the explained variances are presented in relation to Medstack. In this context, it should be added that the boxplot on the far right shows the results without additional information.

and/or DO 4 (cf. Fig. 2.3).

The Late Glacial (**15.4 – 14.2 cal ka BP**) is characterized by a spread of pioneer plants and an increase in Pistacia and Olea europaea (Miebach et al., 2019). This leads to a return to higher temperatures, which is also evident in our reconstruction. The Koeppen-Geiger classification returns to the hot Mediterranean climate Csa with temperatures above 15 °C and P_{ANN} of 480 mm. This variation is also marked in the isotopic references and is called BA interstadial or DO 1 (Rasmussen et al., 2014). All in all, the similarities between the WPS patterns in Fig. 2.3 and the WPS of the reconstructed annual precipitation for this period in Fig. 3.26 (c) are quite close. On the one hand, DO 1 is accompanied by significant power, but DO 2 is somewhat more difficult to detect, and DO 3 – 4 is again more noticeable, although significance is lacking in our reconstruction.

Sensitive study The final reconstruction contains the pollen information from the core, without the AP/NAP curve. Isotopes used are not only those of a speleothem, but also those of planktonic foraminifera from the Mediterranean Sea. In contrast, the reconstruction represented by the blue line in Fig. 3.25 does not take into account the additional information from the speleothem. This results in explained variances that vary from almost 0 to 0.65 with a mean of about 0.3 (see Fig. 3.26 (b)).

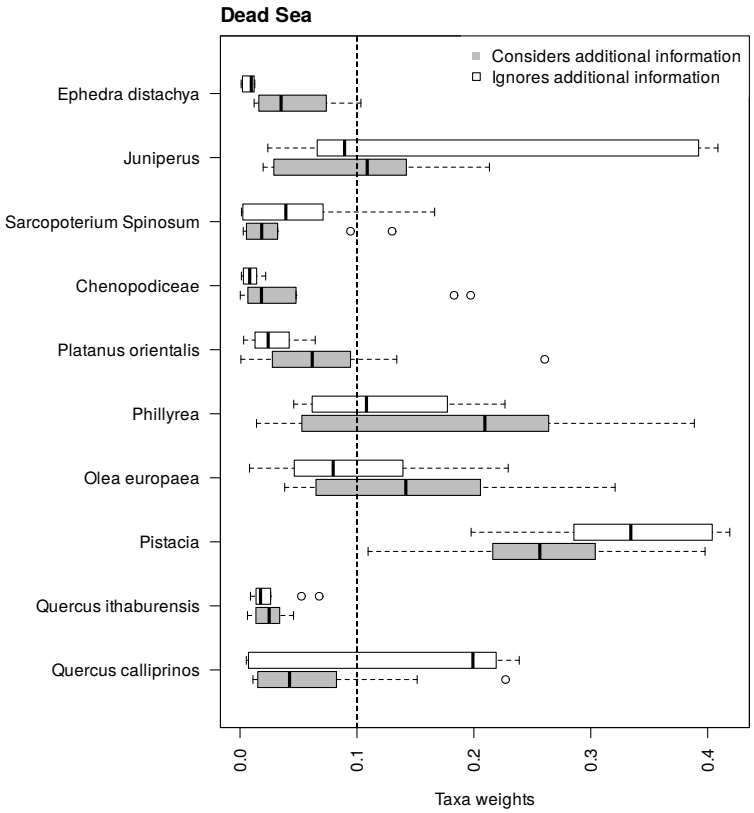


Figure 3.27: As Fig. 3.23, but for the Dead Sea.

Since the Pareto calculations in this case correspond to the highest values of a single variable, the explained variances with respect to Medstack range from 0.55 to 0.65.

By including the additional information, the Pareto optimum involves a compromise between these two proxies. Thus, R^2 from Medstack ranges from 0.4 to 0.6 and is slightly lower than those without additional information. The explained variance related to the Soreq cave is much higher, ranging from $0.6 - 0.65$ in summary. These higher values lead to the main differences in Fig. 3.26 (c) and (d). Around 50 cal ka BP, the WPS without additional information shows high significant power for periods around 16 ka. These higher values lead to the main difference from Fig. 3.26 (c) to (d). For ages around 50 cal ka BP and periods around 16 ka the WPS without additional information reveals high significant power. Instead, the WPS with additional information follows the spectral behaviour of the Soreq cave.

The main difference when looking at the reconstructions is the amplitude of the two variables over the entire period. The maximum values of the blue lines are relatively similar, while the minima differ significantly in some areas. Here they are higher for both variables, leading to the largest differences during the coldest periods, namely MIS 6 and MIS 2. In addition, the fluctuations during the Heinrich events are not as strong as in the case of the inclusion of isotopes from the Soreq cave.

All in all, the reconstruction with both information agrees better with the qualitative climate reconstruction of Chen and Litt, 2018 and Miebach et al., 2019 than when only the Medstack reference is used. This is also confirmed by the fact that the maximum of T_{ANN} in the blue line is not during the LIG optimum, but just after the onset of MIS 5e. Looking at the posterior taxa weights for both models in Fig. 3.27, it is noticeable that, with respect to oaks, Quercus calliprinos is much more weighted than for the case without additional information. This taxon has the highest pollen percentages in cooler periods, resulting in the differences described above. Phillyrea, Olea, and Pistacia, however, occur mainly during early MIS 5. These taxa have well mixed weights for the case with additional information and a maximum mean weight for pistachio. Although this is also the case for the reconstruction without additional information, the other important taxa for early MIS 5 are much less weighted. This results in the shifted LIG optimum when isotopic information from the Soreq cave are neglected.

Summary and conclusion In this section we describe a climate reconstruction of annual temperature and precipitation in the Dead Sea region over a period of about $14.5 - 147$ cal ka BP. Isotopic information from Soreq Cave and the Mediterranean Sea are used as a reference. Although no a priori climate anchor points are used for the reconstruction period, the algorithm provides reasonable climate values for both variables. The reconstructed T_{ANN} for the LIG optimum match those of En Gedi during the Middle Holocene in Sect. 3.3.1. In comparison, P_{ANN} is slightly higher than today. The corresponding Koeppen-Geiger approximations, indicating a hot Mediterranean climate, hold for most of the reconstruction period. Exceptions are the late Penultimate Glacial (late MIS 6) and the Late

Glacial (MIS 2). During these periods, the classification transitions to a cold steppe climate caused by a decrease in T_{ANN} and P_{ANN}. Nevertheless, the lowest climate averages of these glacial conditions result in occurrence probabilities $\mathbb{P}(\boldsymbol{P}|\boldsymbol{C}_{\boldsymbol{Glacial}})$ for Quercus ithaburensis and Quercus calliprinos of about 45 % and 50 %, respectively. If the isotopic information from Soreq Cave is not taken into account, the reconstruction shows higher values for both variables during the cold periods. As a result, the corresponding Koeppen-Geiger classifications remain in the Mediterranean climate. This leads to the conclusion that a compromise between the two isotopic sources (Medstack and Soreq Cave) results in a climate reconstruction that is more consistent with what Chen and Litt, 2018 and Miebach et al., 2019 describe.

3.3.2 Lake Kinneret

Lake Kinneret, also known as the Sea of Galilee or Lake Tiberias, is located about 150 km north of the Dead Sea in the Jordan Valley. The water level of this freshwater lake is about 212 m bmsl and is mainly fed by the Jordan River (Schiebel and Litt, 2018). The climate diagram in Fig. 3.28 of the grid point closest to the Sea of Galilee shows a hot Mediterranean climate Csa. The mean annual temperature in this region is about 18.5 °C, and the average annual precipitation is around 470 mm. Summers are dry and hot, while winters have a mild and humid climate. Most rainfall in this region occurs between October and April. The unique orography of the Jordan Valley results in a strong precipitation and temperature gradient. Lake Kinneret thus lies on the southern edge of the

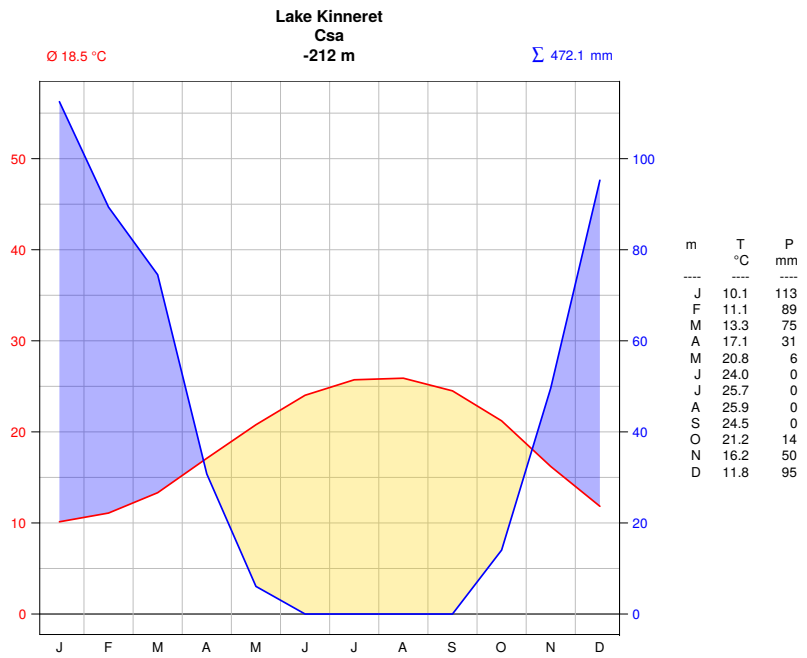


Figure 3.28: As Fig. 3.18, but for Lake Kinneret.

Mediterranean climate and borders the Irano-Turanian steppe (Miebach et al., 2019). This vegetation gradient indicates a relatively high climatic variability also during the Holocene. In the following sections we present such a quantitative climate reconstruction.

Introduction

In 2010 AD, a sediment composite core of 17, 8 m was recovered during a drilling campaign in the Sea of Galilee. The corresponding vegetational data are described in Schiebel and Litt, 2018 and covers approximately the last 9,000 years. Later, this palynological dataset was supplemented by additional samples to increase the temporal resolution (Thomas Litt, and Andrea Miebach, personal communucation). During the reconstruction process, the Koeppen-Geiger classification shown in Fig. 3.28, is used as an anchor for the most recent time slice. The problem of increasing human influence after about 6.5 cal ka BP is treated in the same way as for the Dead Sea in Sect. 3.3.1.

Several references were used for the reconstructions of the Dead Sea region. On the one hand, we include AP/NAP from the core and isotope data from Soreq Cave for a reconstruction during the Holocene. On the other hand, the AP/NAP curve was replaced by isotopic information from the Mediterranean Sea (Medstack) for the period from the Penultimate to the Last Glacial. Now all three proxy information are included, which are shown in Fig. 3.29.

The Bayesian age-depth model of this sediment core is presented in more detail in Miebach et al., 2022. It mainly includes ^{14}C measurements from terrestrial plants, sedimentary charcoal remains, and reservoir-corrected data from bulk sediments.

Reconstruction

In the following we describe the final reconstruction shown in Fig. 3.30. Panel (a) shows the reconstructed annual temperature T_{ANN} in $^{\circ}C$ and (b) the annual precipitation P_{ANN} in mm/a. At the beginning of the explanations we focus on the solid black lines. Later, we will work out the difference between these reconstructions and others whose medians are shown as blue lines. Schiebel and Litt, 2018 divide vegetation history into different ages, which is also used in this work.

The period **9 – 7 cal ka BP** can be associated mainly with the Pottery Neolithic (Schiebel, 2013). The vegetation is described in Schiebel and Litt, 2018 with a strong influence of steppe vegetation in the catchment area of Lake Kinneret. They conclude that this is due to increasing drought, which is confirmed by our reconstruction in Sect. 3.3.1, the lake levels of Lake Kinneret (Hazan et al., 2005) and the Dead Sea (Migowski et al., 2006). Furthermore, Miebach et al., 2022 infers a weak cooling trend and precipitation decrease during 7.8 – 7 cal ka BP from carbon isotope signals of the Sea of Galilee. These descriptions are consistent with our new climate reconstruction in Fig. 3.30. Panel (a) shows a slightly decreasing trend by 0.5 $^{\circ}C$ after 8 cal ka BP. This underlines the assumption that Medstack can be correlated with T_{ANN}. Although the temporal resolution of this proxy is quite low, a

corresponding increase can also be seen here, which is detected by the reconstruction algorithm. The average temperature in these years is about 18.5 °C. In (b) we see the lowest precipitation values of

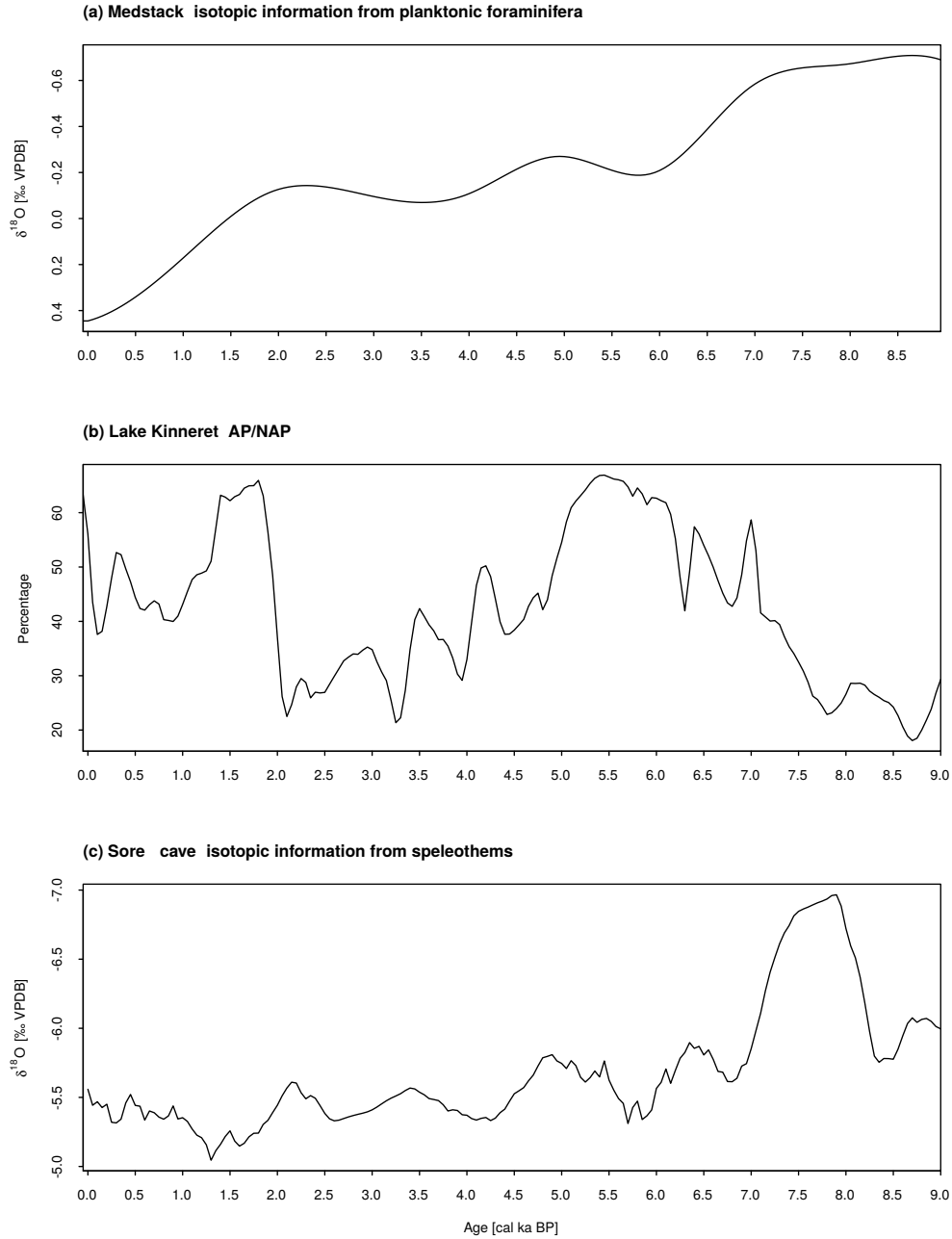


Figure 3.29: Reference curves based on (a) isotopic information from planktonic foraminifera from stacked marine cores of the Mediterranean Sea (Colleoni et al., 2012), (b) on total terrestrial pollen sums: arboreal and non-arboreal pollen taxa from Lake Kinneret, and (c) on isotopic data from speleothems from Soreq Cave (Bar-Matthews et al., 2003).

the entire reconstruction period with mean values around 400 mm and a local maximum around 8 cal ka BP. The approximate Koeppen-Geiger classification remains in the hot Mediterranean climate Csa during the last 9,000 years. Around 7 cal ka BP, an increase in P_{ANN} to values of about 550 mm is observed, with the annual temperature remaining relatively stable.

The beginning of the period **7 – 5 cal ka BP** is accompanied by an increase in Olea europaea. Schiebel and Litt, 2018 assume climate change towards more rainfall, which is also confirmed by the levels of Lake Kinneret and the Dead Sea. The isotopic data from Lake Kinneret also indicate an increase in precipitation and a stabilization of temperature during 7 – 6.6 cal ka BP. Likewise,

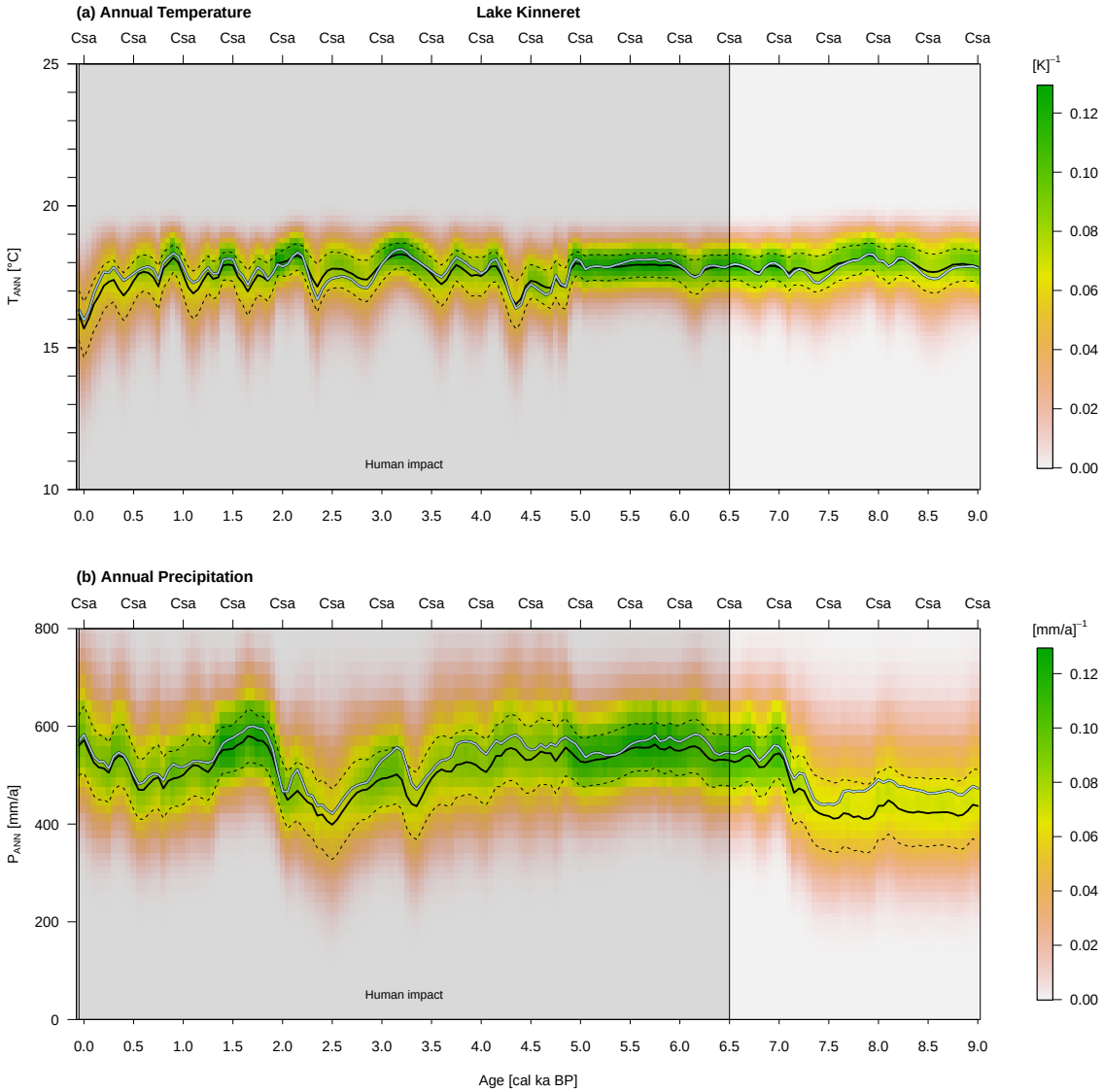


Figure 3.30: As Fig. 3.20, but for Lake Kinneret.

in our reconstruction from the Dead Sea in Sect. 3.3.1, a slight increase in P_{ANN} can be detected during 7 cal ka BP. In relation to the current reconstruction, we can see that these changes are also depicted. For both climate variables, this period shows the most stable conditions with precipitation about 550 mm and temperatures surrounding 18 °C. After the transition from the Pottery Neolithic to the Chalcolithic, ca. 6.5 cal ka BP, human influence is dealt with as described in Sect. 3.3.1. During the Chalcolithic, precipitation shows a local maximum, which decreases after about 5.5 ka BP. The

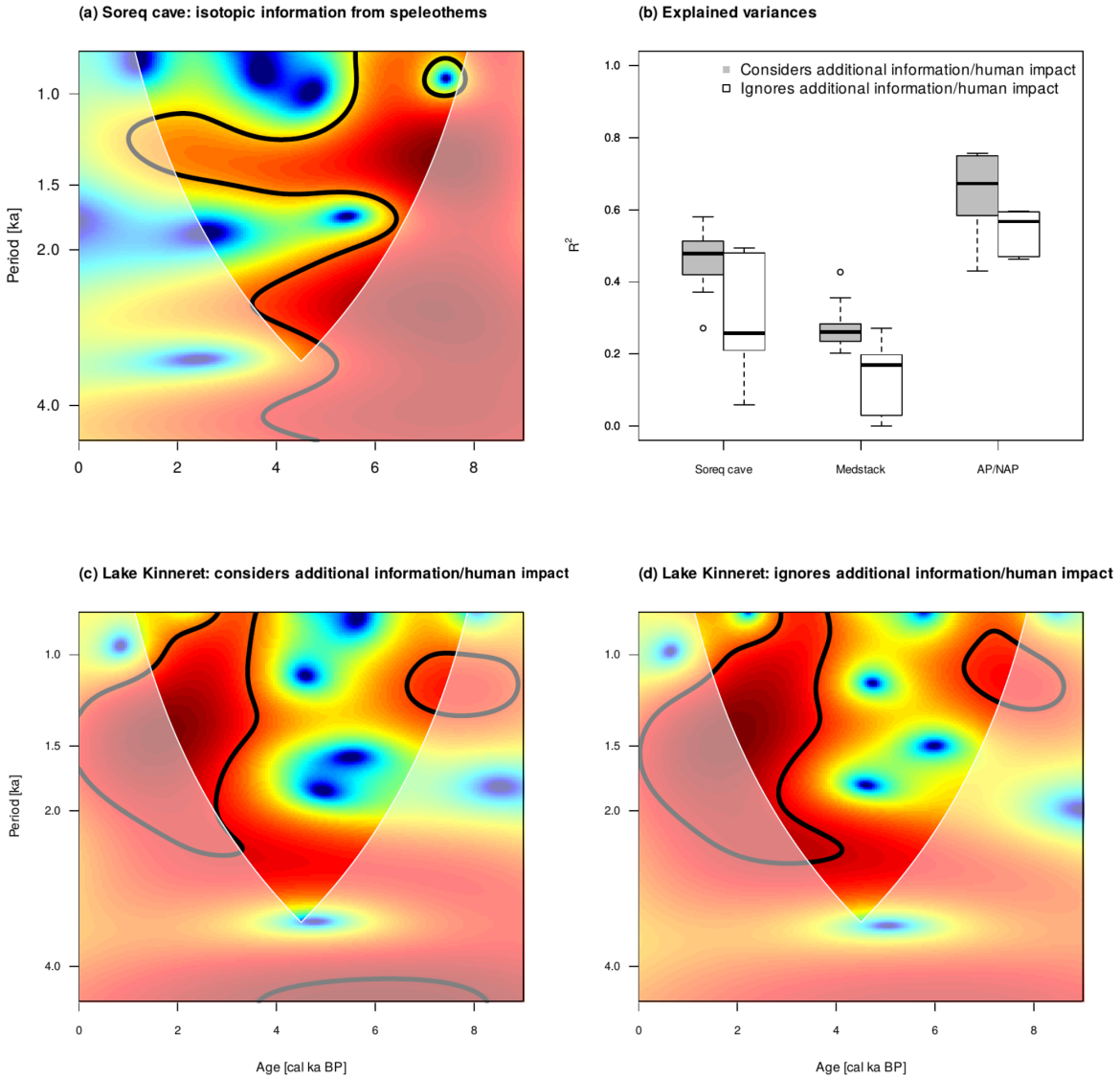


Figure 3.31: As Fig. 3.22, but for Lake Kinneret. Panel (b) is expanded by the explained variances between the medians of T_{ann} and Medstack. Moreover, R^2 in terms of AP/NAP is not divided based on human impact. Instead, only the results of the entire reconstruction period are presented.

second variation is not included in the AP/NAP intercept, but in the isotope data from the Soreq cave. Such behaviour is also described in the Dead Sea reconstruction and could be related to the transition from the Chalcolithic to the Early Bronze Age (Schiebel, 2013).

The Early Bronze Age to Iron Age within **5 – 2.3 cal ka BP** reflects not only human-induced but also climatically driven vegetation changes. On the one hand, Schiebel and Litt, 2018 describe the end of olive cultivation around 5 cal ka BP as a human influence, as this fluctuation does not match the high water levels of Lake Kinneret and the Dead Sea. On the other hand, the decrease in oak pollen by 4 and 3.3 cal ka BP could be related to the 4.2 and 3.2 drought events. Our climate reconstruction shows a slight increase in precipitation and a decrease in temperature from 18 °C to about 16.5 °C for the period 5 – 4 cal ka BP. Here, both isotopic references affect the reconstruction method, with Medstack and Soreq Cave showing an increasing trend leading to lower T_{ANN} and higher P_{ANN}. Accordingly, the climate change to lower precipitation and higher temperatures around 4 ka BP could be related to the transition from the Early to the Middle Bronze Age. The second and larger climate change during 3.3 ka BP might be related to the collapse of the Late Bronze Age (Schiebel and Litt, 2018). Due to the higher resolution of the core samples, we can detect further climate variations. The Iron Age in the Near East lasted from about 3.1 – 2.5 cal ka BP (Schiebel, 2013). This again corresponds to an increase in precipitation at the beginning and ends in a minmium with values around 400 mm. 2.5 cal ka BP marks the transition from the Iron Age to the Babylonian-Persian period, which lasted about 200 years and is accompanied by an increase in P_{ANN}.

The years from **2.3 – 0 cal ka BP** are marked by the Hellenistic and Roman-Byzantine and the Islamic periods until today. Orland et al., 2009 describe these ages, based on isotopic data from Soreq Cave, at the beginning with high precipitation values. At about 1.3 cal ka BP, they found a minmium in the deposit, which they associate with the transition from the Byzantine to the Islamic period. Our reconstruction coincides to a certain extent with these explanations. Precipitation increases up to about 1.75 cal ka BP and has a local minimum around 1.3 cal a BP. However, this is not as pronounced as in the Dead Sea reconstruction, and the reconstructed P_{ANN} of Lake Kinneret subsequently drops further to 0.5 cal ka BP. This shows how difficult it is to carry out a climate reconstruction based on vegetation data, which can be relatively easily manipulated by human activities. Finally, both reconstructed climate variables approach the most recent climate measurements for the last time interval.

Sensitivity study

Looking at the gray boxplots in Fig. 3.31 (b) we see, in contrast to the corresponding results from the Dead Sea (see Fig. 3.22 (b)), a relatively high explained variance with respect to AP/NAP. This suggests that the different proxies used for this reconstruction are more consistent with each other. Even if the additional information and human influence are ignored, R^2 has a similar median with respect to Medstack as in the other case. Nevertheless, the difference is greatest in Soreq Cave, as shown in Fig. 3.31 (c) and (d). The largest changes are found between 3 – 6 cal ka BP and periods

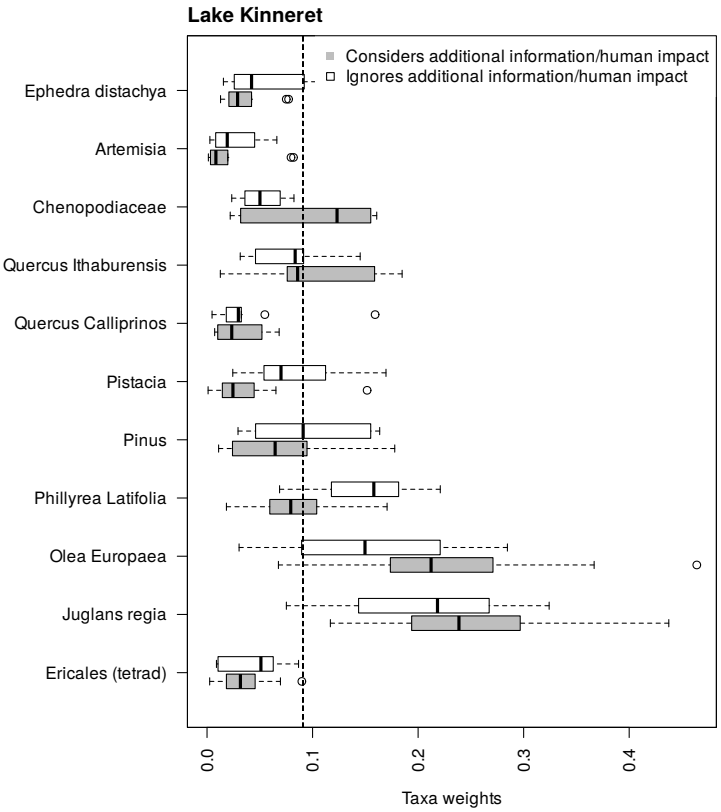


Figure 3.32: As Fig. 3.23, but for Lake Kinneret.

around 1 ka. When the reconstruction method has the information from the Soreq cave, it attempts to remove these higher frequencies and concentrate the branch around the 1.5 ka period (see Fig. 3.31 (a)).

In this context, higher weight is given to Chenopodiaceae (see Fig. 3.32). This has a WPS with relatively high power for such periods around 1.5 ka during these ages (not shown). Some of them are also characteristic members of the Irano-Turanian steppe territory (Schiebel and Litt, 2018) and occur mainly between $9-7$ and $5-2.3$ cal ka BP. During these periods, the blue curve in Fig. 3.30 (b) shows the largest differences with higher values compared to the black line. This is consistent with the relatively low precipitation values of these plants. No such differences can be found with respect to T_{ANN}. This is due to the relatively low variability in mean temperature values within the taxa pool used.

All in all, the differences are not as large as in the previous reconstructions. However, the algorithm again finds the best balance (in terms of R^2 and the climate anchor distributions) between the additional information from the proxies.

Summary and conclusion

The quantitative climate reconstruction of the Lake Kinneret region during the Holocene includes not only paleobotanical data, but also isotopic information from Soreq Cave and the Mediterranean Sea. In the period before 6.6 cal ka BP, the reconstructed climate variables reflect the qualitative climate reconstructions explained in Schiebel and Litt, 2018 and Miebach et al., 2022. Although the final reconstruction is compared only with isotopic information from speleothems and the Mediterranean Sea for ages younger than 6.5 cal ka BP, the results from P_{ANN} are comparable to the lake level of the Sea of Galilee (Hazan et al., 2005). Moreover, the algorithm is able to find climate changes that can be associated with Bond events and known archaeological and cultural changes in the Levant (Schiebel, 2013; Bar-Matthews and Ayalon, 2011; Orland et al., 2009).

Although these changes are not as pronounced as in the Dead Sea reconstruction and the comparatively small difference in terms of sensitivity analysis, we conclude that using the above information is beneficial. However, after 2.5 cal ka BP, human influence cannot be controlled as well as in the case of the Dead Sea. This could be addressed by expanding other proxies for this period, such as lake level or isotopic information from Lake Kinneret. In these cases, both information could be used as further reference curves.

3.3.3 Birkat Ram

Maar Lake Birkat Ram is located about 70 km north of Lake Kinneret in the northern Golan Heights. In contrast to the previously described sites in the Levant, the lake level is about 940 m amsl. Its main source of water is precipitation and local runoff (Schiebel, 2013). The climate diagram in Fig. 3.33 of the grid point closest to Birkat Ram shows a hot Mediterranean climate Csa. The mean annual temperature in this region is about 16°C, and the average annual precipitation is around 620 mm. Summers are dry and warm, while winters have a mild and humid climate. Most rainfall in this region occurs between October and April. Due to the higher elevation and mountainous orography around the lake, the annual temperature is lower and the annual precipitation is higher than at sites in the Jordan Valley. Thus, today Birkat Ram is located in the Mediterranean vegetation zone (Schiebel, 2013). How these present-day features may have changed over the past 30,000 years is discussed in the next sections.

Introduction

In 1999 AD and 2010 AD, two drilling campaigns returned 543 cm and 10, 96 m composite core, respectively. The corresponding vegetation data are described in Schwab et al., 2004 and in Schiebel, 2013 and cover approximately the last 6,500 and 30,0000 years. Schiebel, 2013 find a consistent correlation of pollen ratios as well as magnetic susceptibility signals of the two Birkat-Ram composites. For this reason, and because the shorter core has a higher sample resolution, we merge these two

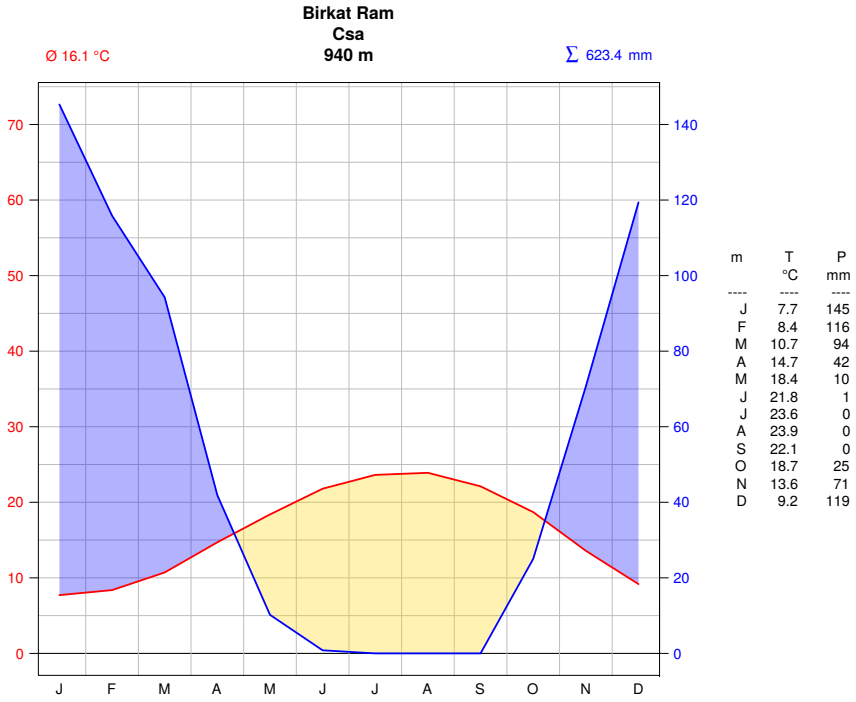


Figure 3.33: As Fig. 3.18, but for Birkat Ram.

palynological data sets. The magnetic susceptibility peaks at 516 cm (Schiebel, 2013) and at 527.7 cm (Schwab et al., 2004) are used to join both pollen spectra. That means, that for ages younger than ca. 6,250 cal a BP, the information of the shorter core are used. The reconstruction method uses the same references as for Lake Kinneret (see Sect. 3.3.2) and are mapped in Fig. 3.34.

According to the linkage of the two sediment cores, the Bayesian age-depth model uses the respective datings. This has already been done in Schiebel, 2013, where the basis of our model is described. Compared to previous age-depth models, the result shows partially discontinuous sedimentation of about 7,000 years between about 10 – 17 cal ka BP (Schiebel, 2013). This zone of low sedimentation is shown in Fig. 3.34 (b) and results in a relatively small deviation of the AP/NAP curve within this period.

Reconstruction

In the following we describe the final reconstruction shown in Fig. 3.35. Panel (a) shows the reconstructed annual temperature T_{ANN} in °C and (b) the annual precipitation P_{ANN} in mm/a. Schiebel, 2013 divide vegetation history into different ages, which is also used in this work.

During the period **30 – 18 cal ka BP** that encompasses the LGM, Schiebel, 2013 describe a steppe-like character in the vegetation composition. They argue that human influence is much less than during the Holocene and conclude that the climate is relatively cold and dry. However, as mentioned

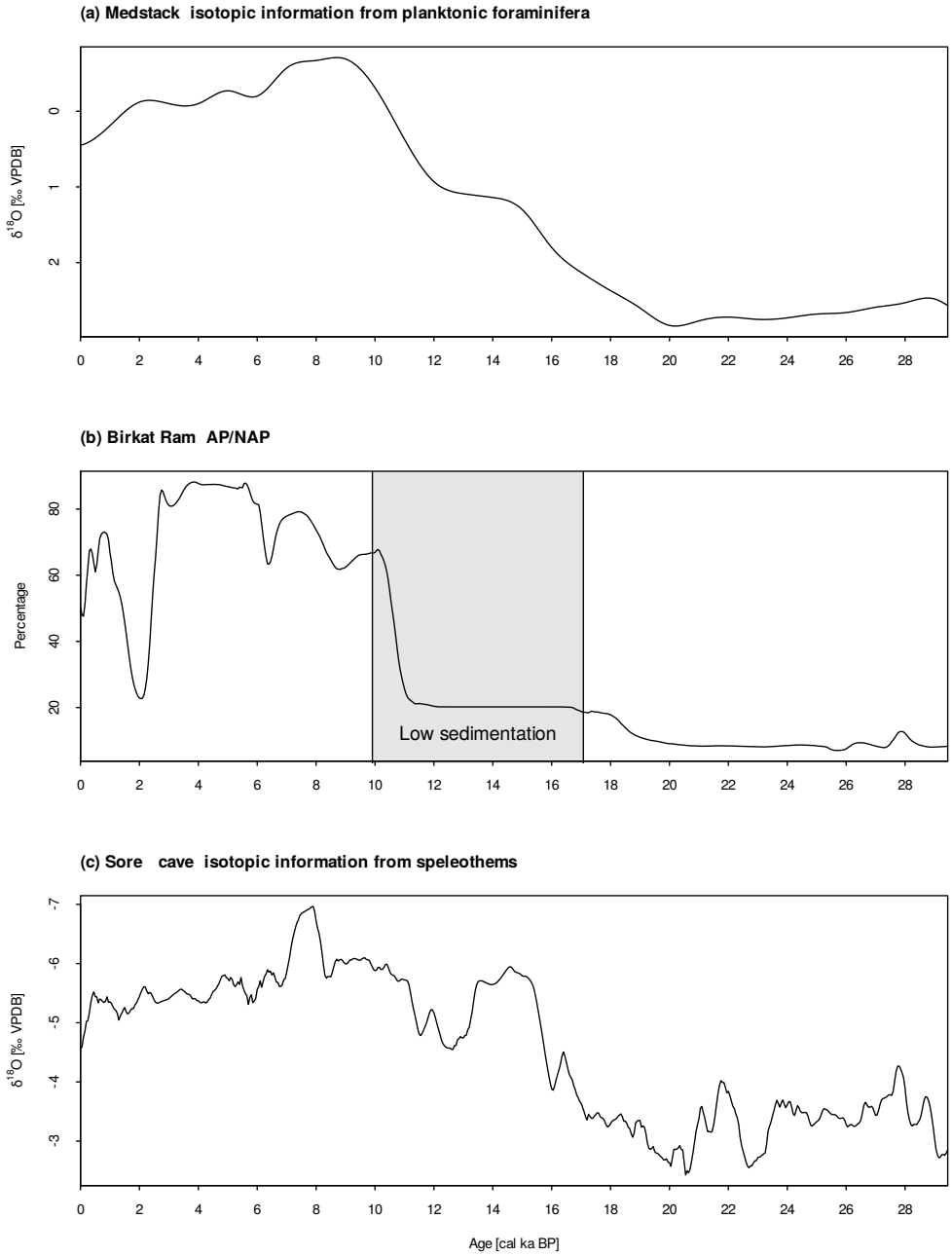


Figure 3.34: As Fig. 3.29, but for Birkat Ram. The grayed out area in (b) marks the zone with low sedimentation rates.

in Sect. 3.3.1, the low evaporation rate could lead to high water levels in Lake Lisan (Schiebel, 2013). Our reconstruction gives an average T_{ANN} of about 8 °C and P_{ANN} of 450 mm. The combination of both variables leads to an approximated Koeppen-Geiger climate of Dfa, indicating a fully humid continental climate with hot summers. For comparison, some central parts of the prairie plains in

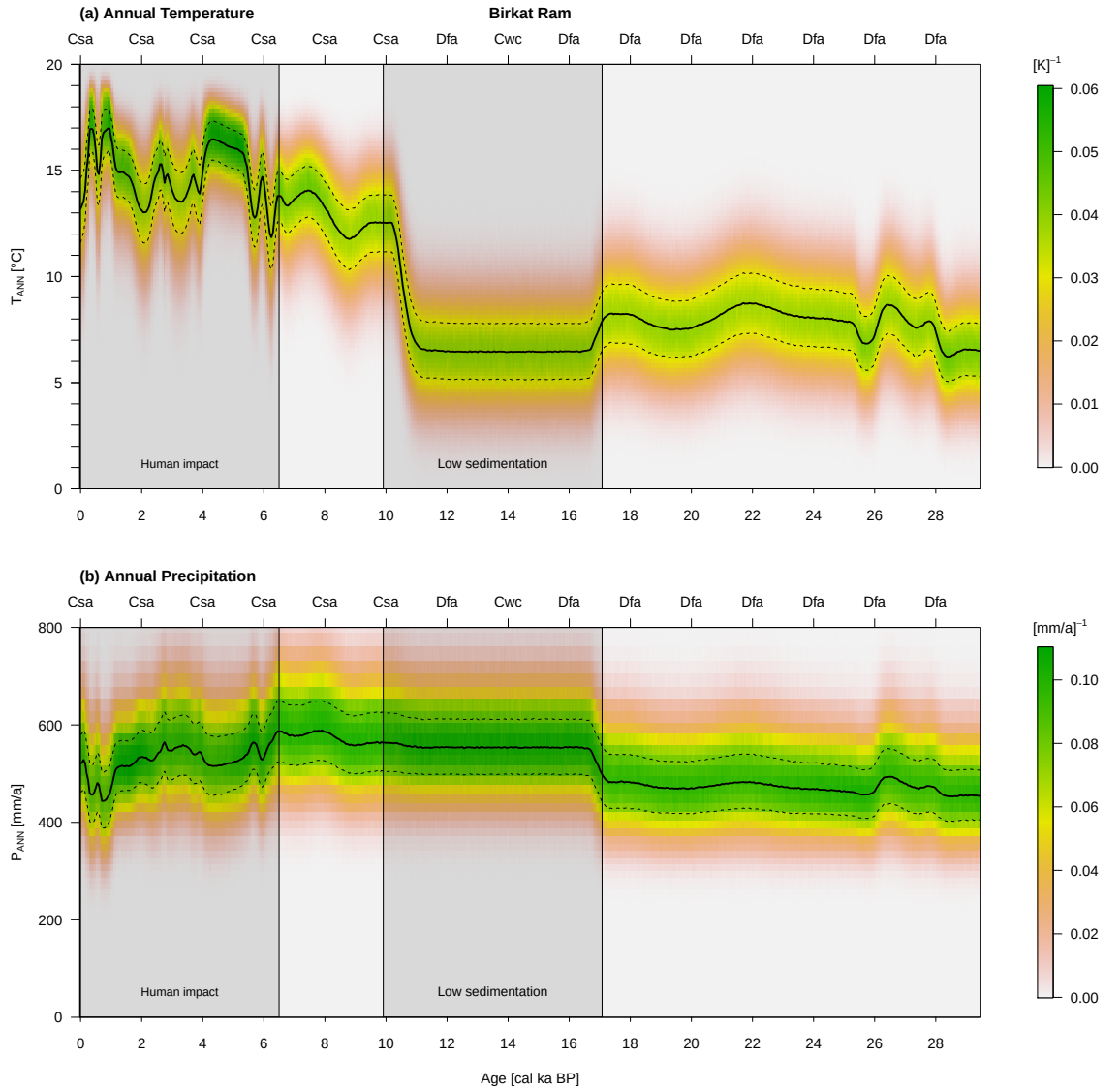


Figure 3.35: As Fig. 3.20, but for Birkat Ram. The grayed out areas mark the zone with low sedimentation rates and human influence. Since no sensitivity study is performed this time, no blue lines are shown.

the USA today have a similar climate and elevation that was dominated by the natural vegetation of the grass-prairie mixture (Weaver, 1965). The corresponding Dead Sea climate in Fig. 3.25 shows warmer temperatures but similar precipitation for this period. Between 30 – 25 cal ka BP, some fluctuations are observed in both variables. In Fig. 3.36 (b) the WPS shows a high significant power with a period of about 1.5 ka. This is consistent with Dansgaard-Oeschgar events 3 and 4 occurring between Heinrich stages 3 and 2 (cf. Fig. 2.3). In the WPS of Soreq Cave we see corresponding periods for ages older than 28 cal ka BP. Although the isotopic information from Greenland is not

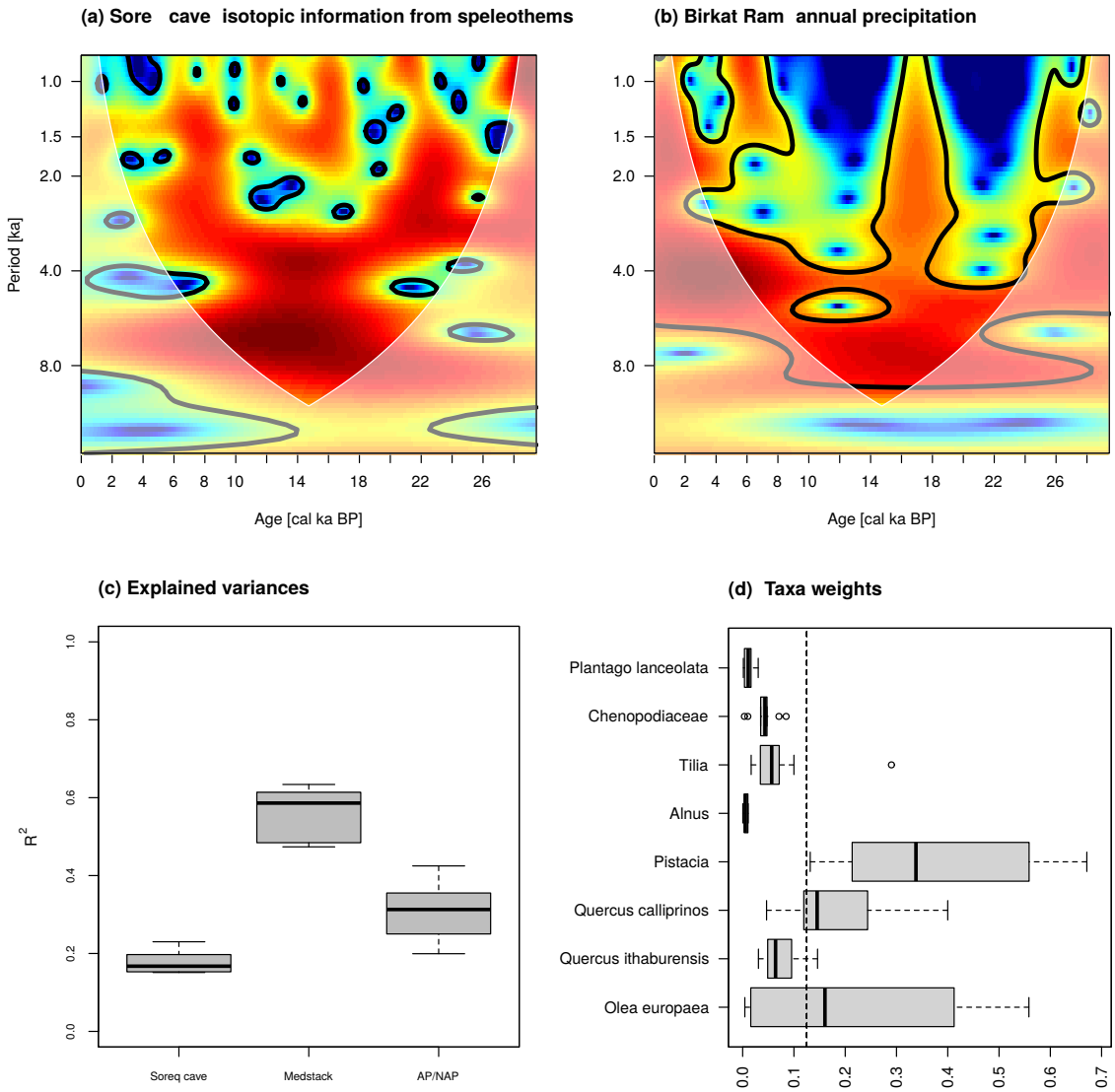


Figure 3.36: Wavelet power spectrum of (a) isotopic information from speleothems from Soreq Cave (Bar-Matthews et al., 2003) and (b) the reconstructed annual precipitation. Panel (c) shows the explained variances between the selected reconstructed medians from P_{ANN} and the patterns from Soreq Cave and the AP/NAP reference curve. The corresponding medians from T_{ANN} are compared with isotopic information from Medstack. In (d), the posterior taxa weights of the selected reconstructions are shown as boxplots and the prior weight is represented by the dashed line.

known to the algorithm, the final result is closer to it than to the data from Soreq Cave. It seems that there are corresponding features in the palynological data, as the coarse resolution of Medstack cannot capture such high-frequency variations.

Between **17 – 12.9 cal ka BP** the lake level was so low that Schiebel, 2013 suggest a possible

desiccation of Birkat Ram during the Deglaciation period. They noted an increased amount of precipitation during DO 1, possibly superimposed by increased evaporation/temperature. This could explain the discrepancy between the low lake level and the higher precipitation. Due to the vegetation gap during the Late Glacial, we cannot draw any conclusions about the reconstructed climate within the grayed-out area.

With the beginning of the Holocene in the period **11.7 – 6.5 cal ka BP**, Schiebel, 2013 describe the strongest change in pollen composition. They note a decline in Irano-Turanian plant populations and an increase in Mediterranean vegetation. From this they conclude an increase in temperature and precipitation. Around 8.7 cal ka BP, a decline in Mediterranean vegetation is noted, which they associate with the 8.2 event, accompanied by colder and drier climatic conditions. In contrast to the Dead Sea, a rapid recovery of the AP is observed after this event, as shown in Fig. 3.34 (b). Our reconstruction fits these explanations quite well. At the beginning of this period, temperatures are around 13 °C and precipitation is about 550 mm. The Koeppen-Geiger classification transitions from Dfa to the hot Mediterranean climate Csa, which remains in this category throughout the Holocene. Thus, during the 8.2 event, T_{ANN} dropped by only 1 °C, and a decrease in precipitation is hardly noticeable. After that, both variables increase to 14 °C and 600 mm. A look at the WPS in Fig. 3.36 (a) shows that Soreq Cave has significant periods of about 1.5 ka during the Holocene. With the onset of the 8.2 event, a similar pattern occurs in the WPS of P_{ANN}, which can be related with some Bond events. Compared to Lake Kinneret and the Dead Sea, it is difficult to determine what fluctuations might be associated with corresponding events.

As shown in the previous reconstructions of the Levant, the period **6.5 – 0 cal ka BP** is treated with a non-negligible human impact on the vegetation (Schiebel, 2013). It is noticeable that the reconstructed precipitation no longer follows the AP/NAP reference. The sharp decline of AP/NAP due to the deforestation of Quercus ithaburensis during the Hellenistic period (Schiebel, 2013) after ca. 2.3 cal ka BP is therefore not reconstructed in P_{ANN}. Instead, the focus is on the pattern from Soreq Cave and the curve from Medstack, which can be seen in the explained variances in Fig. 3.36 (c). The reconstruction algorithm is able to find higher values in terms of AP/NAP (not shown) than the specified range between 0.2 – 0.4. In contrast, the explained variances of Medstack are higher and those of Soreq Cave are lower. All in all, the best combination (Pareto optimum) between these variables and the climate anchor points emerges again. The core thus contains enough variability to generate such explained variances. To achieve this, the algorithm weights the oaks, olives, and pistachios most heavily (see Fig. 3.36 (d)), with the exception of Quercus ithaburensis, which is the main component of the AP ratio (Schiebel, 2013). The decrease in annual precipitation and the simultaneous increase in temperature between 1.3 – 0.5 cal ka BP is due to the reoccupation of vacant areas by evergreen oaks and pistachios during the Islamic period (Schiebel, 2013). Finally, the reconstructed variables approach the most recent climate anchor point (Csa) for the most recent time slice.

Comparison to older climate reconstructions

In Neumann et al., 2007 a reconstruction of the last ca. 6,500 years based on the older, higher resolution core is presented. In contrast, Thoma, 2017 presents a reconstruction of the last 30,000 years based on the newer, coarser resolution core. The techniques used in both studies are slightly different. While Neumann et al., 2007 use the indicator taxa method, Thoma, 2017 extend it with additional biome information.

The two biomes used in the latter case are the Mediterranean and the Irano-Turanian. On the one hand, the main difference between them in terms of annual precipitation is that the Mediterranean biome has higher values. The Irano-Turanian biome, on the other hand, has lower winter and higher summer temperatures. The biome ratio in Thoma, 2017 resembles the AP/NAP curve, with AP reflecting the Mediterranean biome. This leads to an overall reconstruction that also takes into account human-induced changes during the Holocene. In addition, the summer temperature during the Last Glacial is about 28 °C and the winter temperature is around 10 °C. This is in contrast to the remarks of Schiebel, 2013. In this context, the addition of another biome, such as the Irano-Turanian steppe forest (Litt et al., 2021), could help reconstruct more reasonable values during glacial periods.

In contrast, Neumann et al., 2007 present a reconstruction that shows values for the youngest time slice that are more in line with those in Fig. 3.33. In terms of P_{ANN}, it is similar to our reconstruction, with an overall slight decrease over the last 6,500 years, but with smaller fluctuations than in the reconstructed temperatures. Except for the above-mentioned variation in our reconstruction during $1.3 - 0.5$ cal ka BP, the variables in Neumann et al., 2007 show greater similarity to the variations in the AP/NAP curve derived from Quercus ithaburensis. In our case, this taxon has on average a lower posterior weight compared to the prior weights.

Summary and conclusion

The quantitative climate reconstruction of the Birkat Ram region during the last ca. 30,000 years includes not only paleobotanical data, but also isotopic information from the Soreq Cave and the Mediterranean Sea. In summary, only 8 taxa information contain enough variation that a trade-off between these 3 proxies and 2 climate anchor points can be found. Furthermore, the final result resembles the qualitative reconstruction described in Schiebel, 2013. Thus, the climate classification assumes a Mediterranean climate throughout the Holocene and a continental climate during the Last Glacial.

Furthermore, in the period after 6.5 cal ka BP, most human-induced vegetation changes are not included in the reconstruction. Compared to old quantitative reconstructions, our new algorithm is better able to avoid anthropogenic fluctuations in palaeo-vegetation and to reconstruct a more appropriate climate during the Last Glacial.

3.3.4 Lake Van

Lake Van is located in the East Anatolian High Plateau in a tectonic depression (Litt et al., 2014). It is the fourth largest terminal lake on earth and is surrounded by the Bitlis massif, which reaches an altitude of more than 3500 m amsl (Pickarski et al., 2015b). The lake itself is at an elevation of about 1640 m amsl. The climate diagram in Fig. 3.37 of the grid point closest to Lake Van shows a Mediterranean-influenced continental climate (Dsb) with warm and dry summers. The average annual temperature in this region is about 8.5 °C, and the average annual precipitation is around 550 mm. The highest rainfall occurs in winter and spring and is strongly influenced by the orography. Thus, two vegetation types are found in the vicinity of the site: the Kurdo-Zagrosian oak steppe-forest and the Irano-Turanian steppe (Pickarski et al., 2015a). How these present-day features may have changed over the past 130,000 years is discussed in the next sections.

Introduction

Two drilling campaigns in 2004 and 2009 recovered a single core (VAN04-2) and a composite core from the Ahlat Ridge. The first is described in (Litt et al., 2009) and covers a continuous sequence from the LGM to the present. Part of the chronology is based on varve information. The second covers the last 600,000 years (Litt et al., 2014) and the period from about 10 – 130 cal ka BP is described in Pickarski et al., 2015b and Pickarski et al., 2015a. The corresponding chronology is based on, among

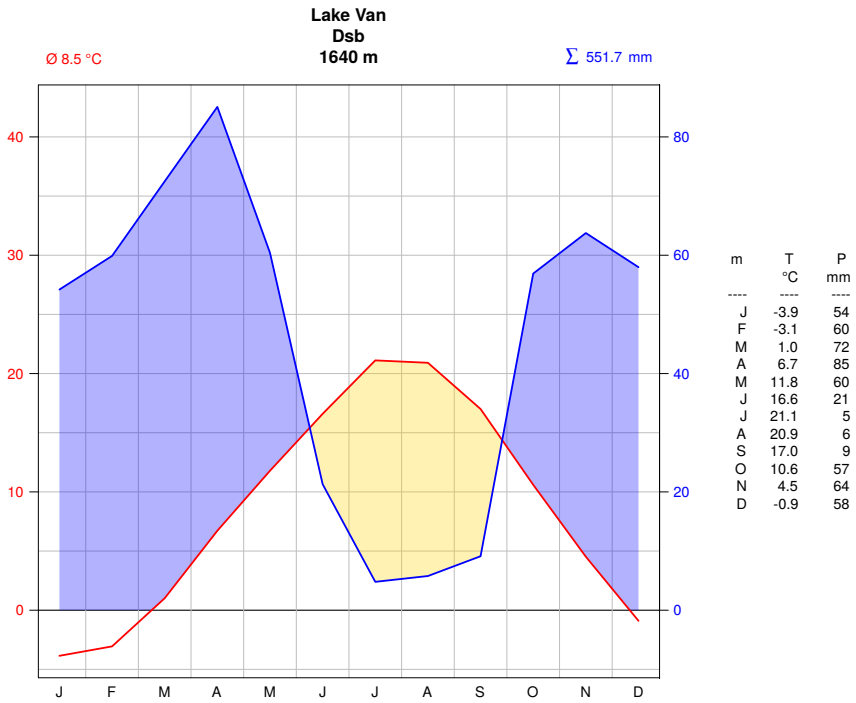


Figure 3.37: As Fig. 3.18, but for Lake Van.

other things, the same varve information as for core VAN04-2 (Stockhecke et al., 2014), so we use the same age-depth model for both cores. For this reason, and because of the proximity of the two sampling sites to each other (Litt et al., 2009), we summarize the corresponding vegetation spectra as for Birkat Ram in Sect. 3.3.3. The linkage point is at about 20 cal ka BP. The result of the combined AP/NAP reference curve in Fig. 3.38 (b) illustrates this procedure. We see a local maximum around 13 cal ka BP and a subsequent local minimum at about 12 cal ka BP. This is consistent with the results presented in Litt et al., 2009. In addition, we note a sharp decrease in percentages by 3.8 cal ka BP where Litt et al., 2009 mention an increase in human activity. For this reconstruction, the information on Pinus was removed from the AP summary. Pickarski et al., 2015b state that this taxon is likely Pinus nigra, and Pickarski et al., 2015a describe its distribution as an indicator of a colder/drier

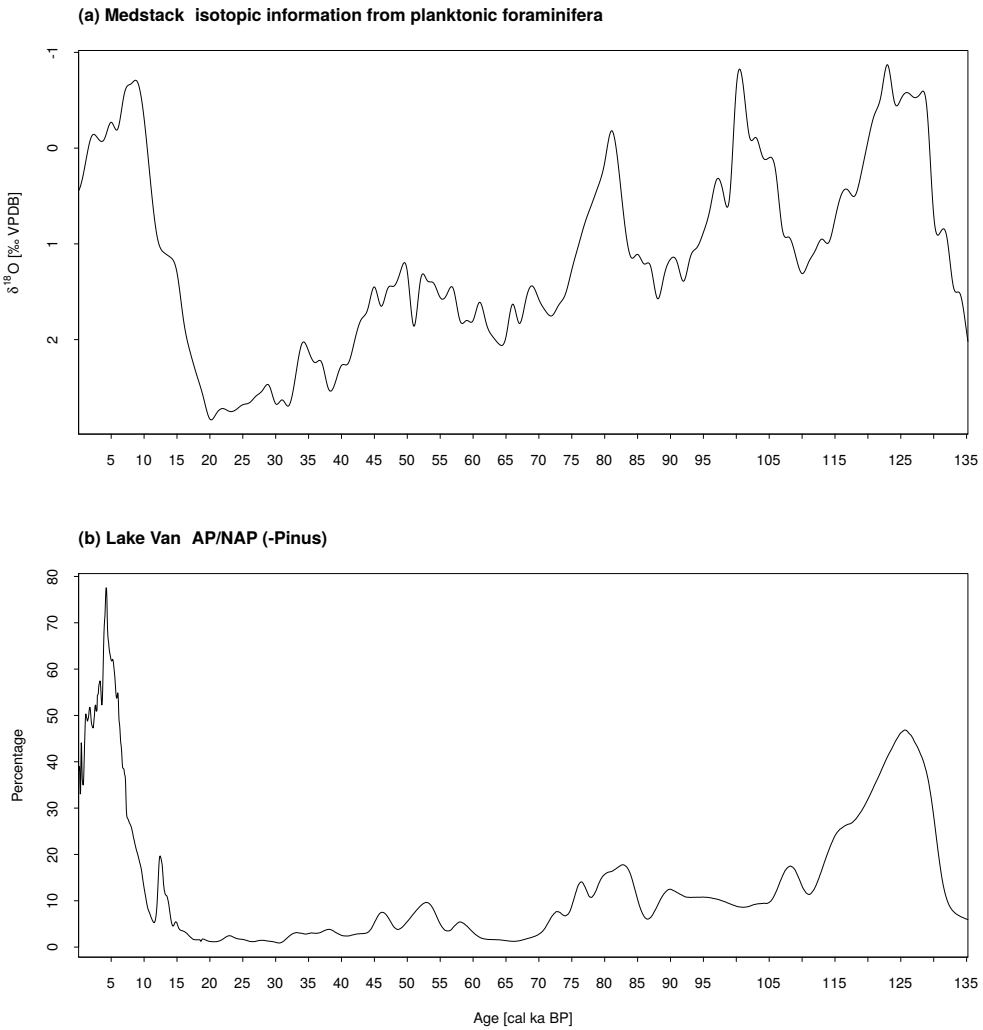


Figure 3.38: As Fig. 3.29, but for Lake Van. Pinus is removed by calculating the percentages of arboreal pollen.

environment. Therefore, the main components of the AP are the temperate taxon deciduous oak, the moisture-demanding thermophilic taxa such as Ulmus and Carpinus betulus, and the frost-sensitive Pistacia (Pickarski et al., 2015b). They also describe each interstadial as a "warm/wet" phase that could be associated with our AP. So we compare this curve not only for annual precipitation as in the Levant, but also for annual temperature.

The relationship between age and depth used in this work is described in more detail in Stockhecke et al., 2014. It is based not only on varve counts, but also on radiocarbon dates and tephra layers. Furthermore, they derive age control points from a correlation of Lake Van paleoclimate records with δ^{18}O ice core records from NGRIP and GLT_{syn} SpeleoAge. The latter is a speleothem-based synthetic Greenland ice-core record. For the last 20,000 years, the reported age errors are relatively small and increase thereafter. If no errors are specified, we use the mean error of the specified uncertainties around these ages. All data up to the first gap around 260 cal ka BP are inserted into the Bacon model. A result of this age-depth transformation can be seen in Fig. 3.38. (b), where we find a much higher age resolution for years younger than 20 cal ka BP, leading to higher frequencies in the time series.

Reconstruction

In the following, we describe the final reconstruction shown in Fig. 3.39. Panel (a) shows the reconstructed annual temperature T_{ANN} in °C and (b) the annual precipitation P_{ANN} in mm/a. Litt et al., 2009, Pickarski et al., 2015b, and Pickarski et al., 2015a divide the corresponding vegetation history into different periods, which are also used in this work.

Pickarski et al., 2015a describe the late Penultimate Glacial from **133.9 – 131.2 cal ka BP** with a dominance of NAP. They conclude cold and dry climatic conditions, which are also reflected in our reconstruction. The average T_{ANN} of about 6.5 °C and P_{ANN} of ca. 400 mm lead to a Koeppen-Geiger classification of continental climate Dsb. At the end of this period, according to the reference curves in Fig. 3.38 (a) and (b), a climate change toward warmer and wetter conditions can be observed.

The pre-temperate phase from **131.2 – 129.1 cal ka BP** shows a decrease in NAP with patches of temperate trees (Pickarski et al., 2015a). This indicates a generally warmer environment with increased precipitation, which is also reconstructed in our study. We see a temperature increase of about 2 °C and a precipitation increase of ca. 200 mm. The corresponding climate classification transitions to a Mediterranean climate with warm summers (Csb) that persisted during the Last Interglacial.

The LIG plateau, which lasted from **129.1 – 115.6 cal ka BP**, is described by Pickarski et al., 2015a as having a distribution of temperate trees and is divided into three vegetation zones. The Quercus-Ulmus zone 129.1 – 127.2 cal ka BP is characterized by a warm-temperate environment with enhanced precipitation. Here our reconstruction shows the highest annual precipitation and warmest annual temperatures during the LIG. The Carpinus zone of 127.2 – 124.1 cal ka BP is described with a slight decrease in temperature but constant humidity. This slight decrease is reconstructed for both variables. During the Pinus zone 124.1 – 115.6 cal ka BP, a spread of drought-adapted and/or

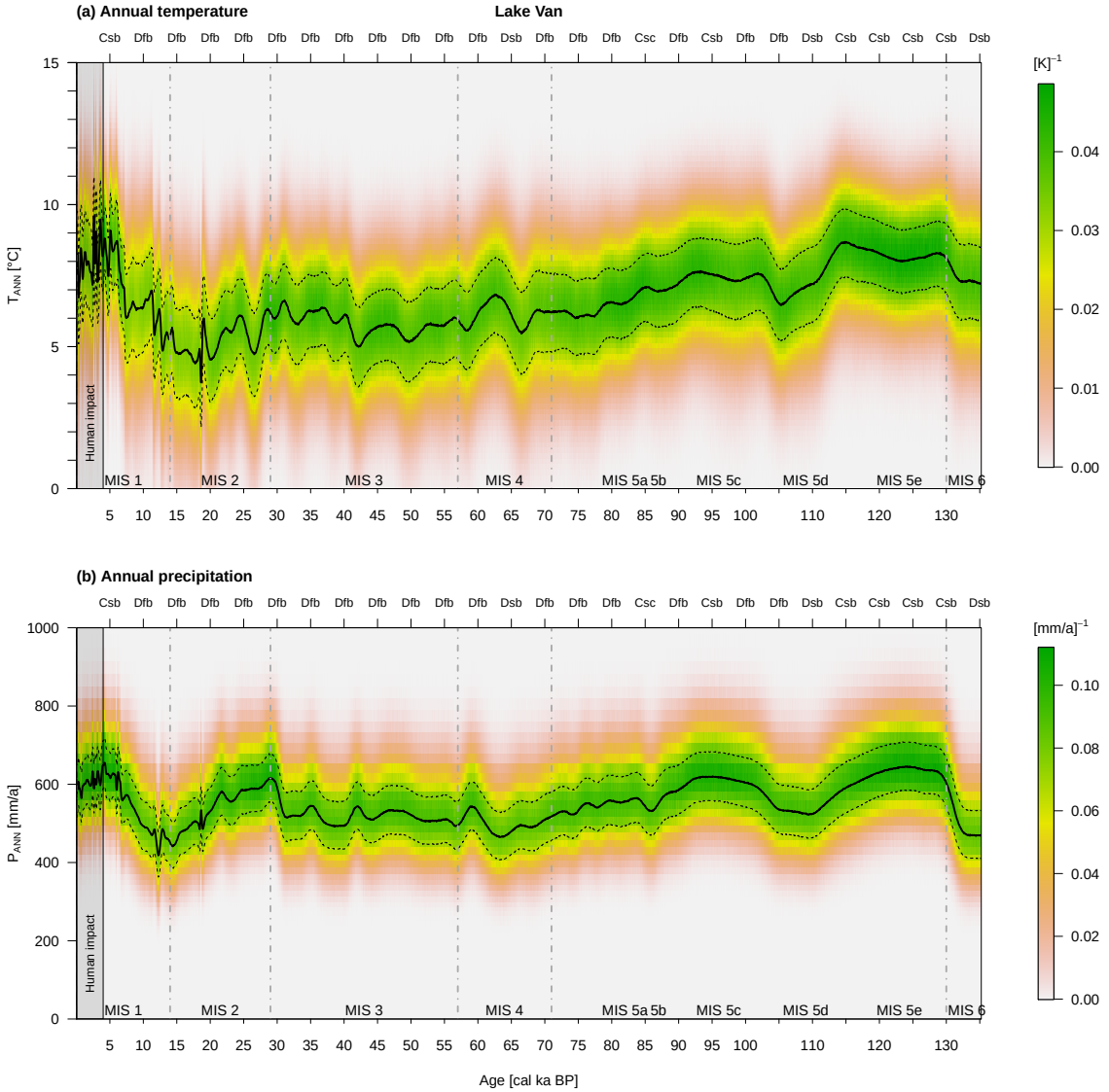


Figure 3.39: As Fig. 3.20, but for Lake Van.

cold-tolerant Pinus is observed, suggesting increasing aridity and cooler temperatures. There is a corresponding successive decline in P_{ANN}, with annual temperature increasing again, but not reaching the level of the Quercus-Ulmus zone. Although both reference curves also describe a decrease in T_{ANN}, the relatively high posterior weight for the warm-temperate Quercus deciduous shown in Fig. 3.40 (b), which still shows comparatively high abundance during this period, could lead to these temperatures.

The post-temperate phase from **115.6 – 111.5 cal ka BP** shows an expansion of NAP, which Pickarski et al., 2015a describe as a quasi-stadial state marking the end of the LIG at about 111.5 cal

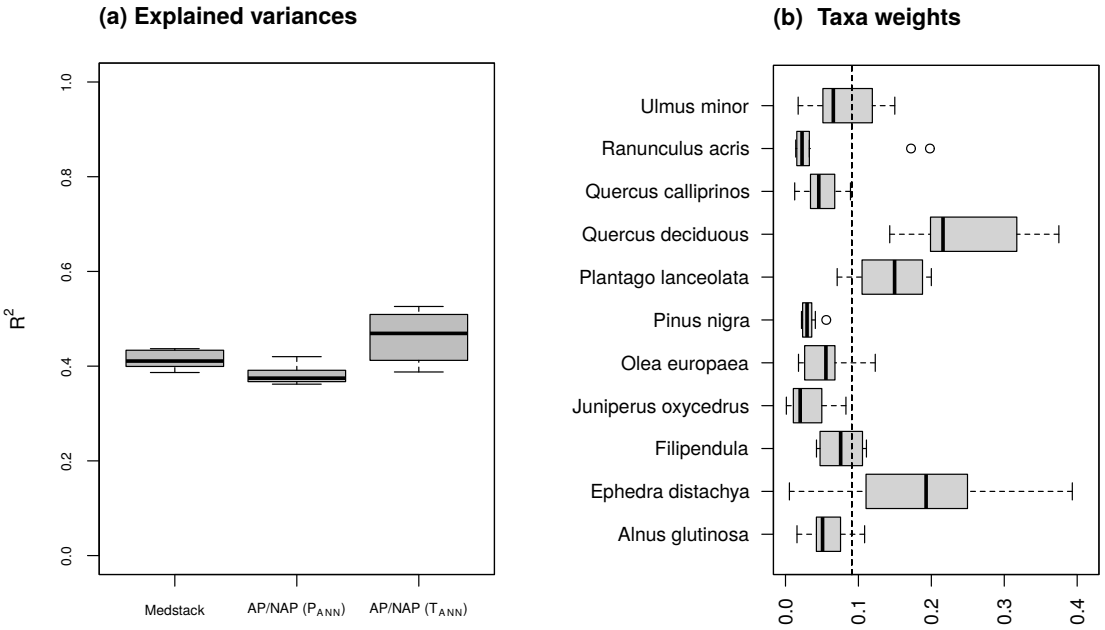


Figure 3.40: Panel (a) shows the explained variance of the posterior samples compared to the Medstack and AP/NAP reference curves. The latter is divided into T_{ANN} and P_{ANN}. Panel (b) depicts the posterior taxa weights.

ka BP. Both the reconstructed temperatures and precipitation are consistent with these explanations, as they each show a decrease. The Koeppen-Geiger classification transitions to the continental climate Dsb.

The following period of **111.5 – 107.8 cal ka BP** could be associated with MIS 5d, also known as the Herning Stadial. Pickarski et al., 2015b describe a significant spread of steppic herbaceous plants and conclude cold and/or dry climatic conditions with pronounced seasonality of precipitation during the stage. This is supported by a reconstructed summer-dry continental climate with annual temperatures around 6.5 °C and annual precipitation of about 420 mm. The relatively wide climatic range of Dsb becomes clear here when we compare the difference of the reconstructed to the current climate values from Fig. 3.37. If we look at the corresponding WPSs in Fig. 3.41, we see a comparatively high power with a period of about 23 ka, lasting at least from 30 – 100 cal ka BP. This could be related to orbital precession, which has a similar period on average and primarily controls the intensity of summer solar radiation (Huybers, 2006). Moreover, in the period of at least 50 – 80 cal ka BP, we see another branch of relatively high power with a period around 41 ka, which could indicate another Milankovic cycle, namely obliquity. Bosmans et al., 2015 mention that this variation also controls solar radiation in summer. So both cycles are connected to this forcing. In Pickarski et al., 2015b, summer solar radiation is used as an additional climate proxy, which often coincides with the vegetation variations

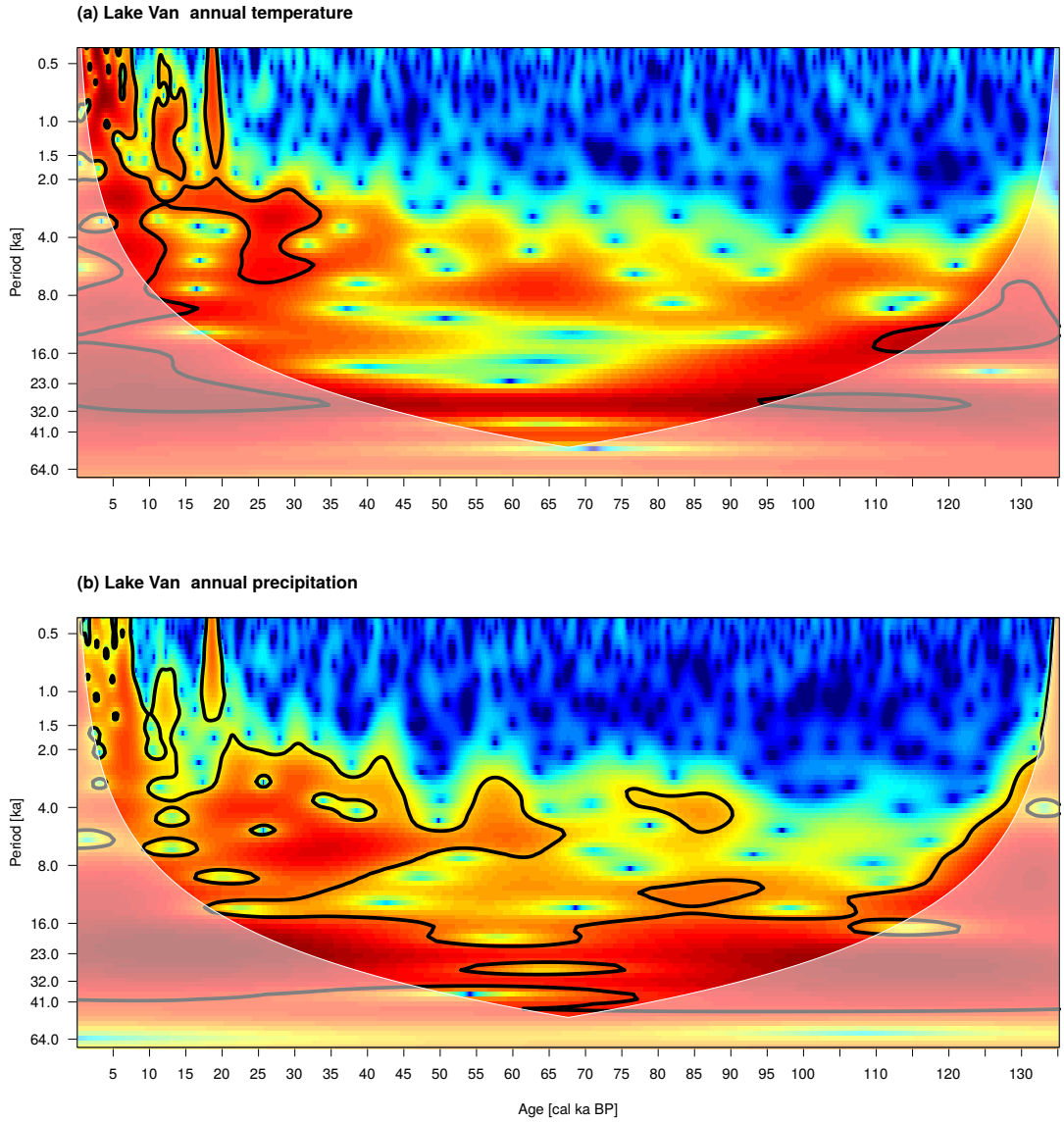


Figure 3.41: Wavelet power spectrum of (a) the median of the annual temperature and (b) the median of the annual precipitation.

of Lake Van. Although no intra-annual climate information is known to the algorithm, we see patterns in this spectral analysis that confirm these descriptions.

After the Herning stadial, the so-called Brørup interstadial (MIS 5c) lasted from ca. **107.8 – 87.3 cal ka BP**. Pickarski et al., 2015b describe it with a dominance of cold and/or summer-dry adapted Pinus. During this period, the reconstruction shows an increase in both variables. Nevertheless, the level of MIS 5e is not reached again, which could be explained by the lower proportion of Quercus deciduous during the Last Glacial. Accordingly, the Koeppen-Geiger classification alternates between

Csb and Dfb. The latter describes a continental climate without a dry season.

The Rederstall stadial of **87.3 – 84.9 cal ka BP** (MIS 5b) behaves like the Herning stadial (Pickarski et al., 2015b) and is accompanied by a reconstructed decline of T_{ANN} and P_{ANN}. The latter shows a local minimum at the end of this period, but the temperature continues to drop. A Mediterranean climate with cold summers (Csc) appears.

MIS 5a, also known as the Odderade interstadial, lasted from about **84.9 – 77.5 cal ka BP** and is accompanied by a slight increase in AP (Pickarski et al., 2015b). They infer an increased moisture supply at the beginning and a decrease in precipitation and/or an increase in evaporation during this period. This interstadial is not as clearly visible in the reconstruction as MIS 5c. Nevertheless, the reconstructed precipitation increases slightly at the beginning of this interstadial, leading to the continental climate Dfb.

Pickarski et al., 2015b describe the period **75 – 28 cal ka BP** with a dominance of Artemisia, Chenopodiaceae and Poaceae. They mention a strong aridification and cooling in Eastern Anatolia after 70 cal ka BP. Our reconstruction shows a relatively stable level in both variables, namely a moderate variation around 6.5 °C and an averaged precipitation of about 450 mm. Although the climate reveals colder temperatures and less precipitation than today, the Koeppen-Geiger estimate remains within Dfb. Pickarski et al., 2015b mention a high variability of the tree population during MIS 3. They associate these alternations of warmer/moist interstadials and cooler/drier stadials with some Dansgaard-Oeschger cycles. Due to the relatively large uncertainty taken into account when transforming depth into age within this period, these fluctuations cannot be detected. This can be verified in Fig. 3.41, where no relatively high power is shown for the typical DO period of about 1.5 ka. We tested a reconstruction with a different age-depth model based on lower age errors for this period. A WPS of these results reveals such high frequencies (not shown), highlighting the importance of accounting for uncertainty in the age-depth transformation. Although the core of Pickarski et al., 2015b and Pickarski et al., 2015a is investigated with a comparatively high resolution, the information of the DO events cannot be resolved after the conversion from depth to age.

MIS 2 from **28 – 14.5 cal ka BP** is described in Pickarski et al., 2015b with cooler and drier conditions compared to MIS 3. The temperature we reconstructed is consistent with this description. However, precipitation increases at the beginning and decreases after about 22 cal ka BP. This is due to Ranunculus acris with comparatively low T_{ANN} but high P_{ANN}. Nevertheless, the amount of precipitation is not greater than in the interstadial MIS 5c. In Litt et al., 2009, the period from circa 20 – 14 cal ka BP that includes the LGM is characterized by cold and semi-desert steppe vegetation. As a result, both climate variables decrease to a minimum of about 5.5 °C and about 400 mm.

The onset of MIS 1 from **14.5 – 11.5 cal ka BP** is indicated by higher pollen values from trees and shrubs (Litt et al., 2009). In both WPSs in Fig. 3.41 we see significant power with periods of about 0.8 – 2 ka. This could include the variations of DO 1 and YD. As far as the climate reconstruction is concerned, both variables show high-frequency fluctuations during their increase. However, the

relatively low amplitudes make it difficult to detect DO 1 and YD.

During the Holocene from **11.5 – 0 cal ka BP**, Litt et al., 2009 mention a sharp increase in moisture. From a growth delay of 3,000 years in deciduous oaks, they infer a dry spring and summer in the early Holocene. This is reconstructed in both variables and we still see a continental climate classification. During the MH (7 – 4 cal ka BP) this transitions to a warm Mediterranean climate with temperatures around 9 °C and around 620 mm. The following period until today is characterized by increased human influence (Litt et al., 2009). We treat this as with the reconstructions before (e.g. section 3.3.1). As a result, temperature and precipitation do not show a strong decrease around 3.8 ka BP as in the AP/NAP curve. The WPS show significant power at higher frequencies during the Holocene, enabled by the relatively small age errors. They also cover the typical period of Bond events around 1.5 ka, but it is difficult to determine which climate fluctuation belongs to which event. Finally, both reconstruction variables of the recent time slices approach the current climate measurements.

Summary and conclusion

The quantitative climate reconstruction of the Lake Van region during the last ca. 130,000 years includes not only paleobotanical data, but also isotopic information from the Mediterranean region. Compared to the reconstructions from the Levant, the AP/NAP values do not contain Pinus components and are correlated with both climate variables. As a result, the posterior taxa weights in Fig. 3.40 (b) show a relatively low weight for Pinus nigra. In contrast, Quercus deciduous and Ulmus minor have mean posterior weights that are higher than the prior values. They are important to determine the warmer and wetter interstadials. On the other hand, Ephedra distachya has the highest mean posterior weight and is used by the algorithm to describe the end of the late Penultimate and late Last Glacial. In summary, only 10 taxa information contain enough variation to find a compromise between these two proxy references and two climate anchor points. This can be checked in Fig. 3.40. (a), where the corresponding explained variances have values between 0.4 – 0.6.

Moreover, the final result resembles quite well the qualitative reconstructions described in Pickarski et al., 2015b; Pickarski et al., 2015a; Litt et al., 2009. Interestingly, the most recent climate classification also occurs at the end of the Penultimate Glacial. Nevertheless, the respective reconstructed climate values are about 1.5 °C and about 200 mm lower than today, which does not contradict the qualitative reconstructions mentioned above. This finally leads to a comparatively low variation of T_{ANN} and a relatively high variability of P_{ANN} throughout the reconstruction period.

3.3.5 Lake Iznik

Lake Iznik is the largest lake in the Marmara region, which lies between the Mediterranean and the Black Sea (Miebach et al., 2016). The lake is surrounded by mountain ridges with peaks between 810 and 1293 m amsl. In Fig. 3.42 we see a climate diagram of the grid point closest to Lake Iznik, describing a warm-summer Mediterranean climate Csb with warm/dry summers and mild/wet winters. The average annual temperature in this region is about 12.2 °C, and the average annual precipitation is 610 mm. Due to the orography, this value varies greatly, which affects the vegetation around Lake Iznik. Litt et al., 2021 describe it with a transition from Mediterranean woodland and mesic Euxinium forest. The latter is adapted to lower temperatures and higher precipitation than Mediterranean vegetation. How these present-day features might have changed over the last 31,000 years is discussed in the next sections.

Introduction

In 2005 AD and 2009 AD, several cores were taken during two drilling campaigns. Miebach et al., 2016 use a composite of some of these cores and investigate the corresponding vegetation history of the last ca. 31,000 years. The proxy references used are shown in Fig. 3.43, which are applied in the same way as in Sect. 3.3.4. For tree pollen, we only consider temperate trees and shrubs (Miebach et al., 2016). As with the reconstruction of the Dead Sea during the Holocene (see Sect. 3.3.1), we use

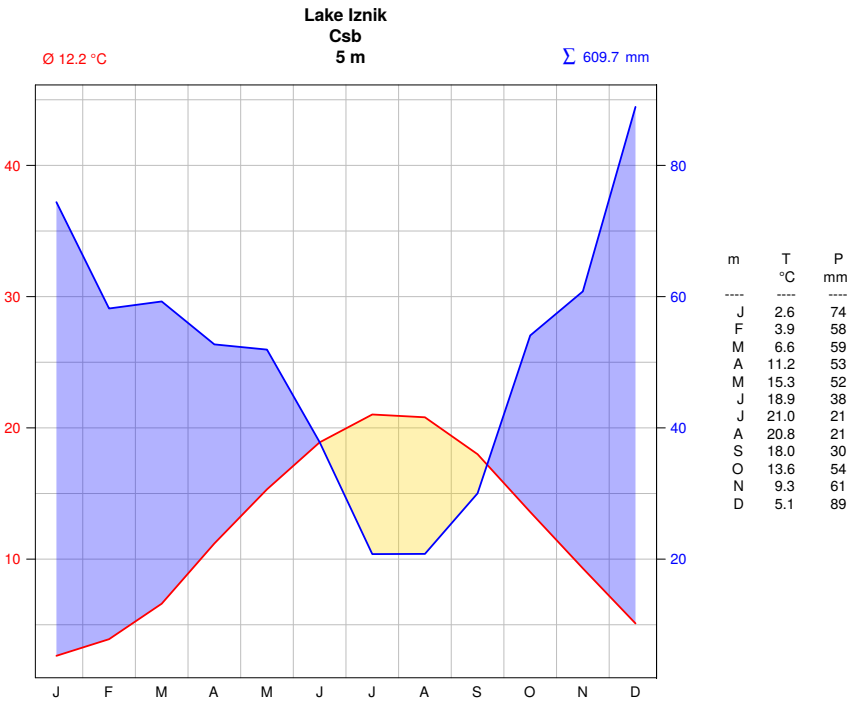


Figure 3.42: As Fig. 3.18, but for Lake Iznik.

the recent Koeppen-Geiger classification Csb as the climate anchor point, although the most recent period is 1400 AD. This assumption is confirmed by ca. 80 % of our local reconstructions over the area depicted in Fig. 3.3, where no change in classifications is apparent in the last 600 years. The marginal probability density distributions of the recent climate anchors are shown here as light green areas in Fig. 3.44.

In contrast to the previous reconstructions, we add an additional climate anchor point for a past period. This can be seen in Fig. 3.44 (b) at 21 cal ka BP, which corresponds to the LGM. Miebach et al., 2016 mention that precipitation during this period was relatively low and can be compared to that of the steppe in central Anatolia. In Fig. 3.17 (c) we see some grid points in central Anatolia indicating a cold steppe climate (BSk). Therefore, these are used to create the PDF of the annual

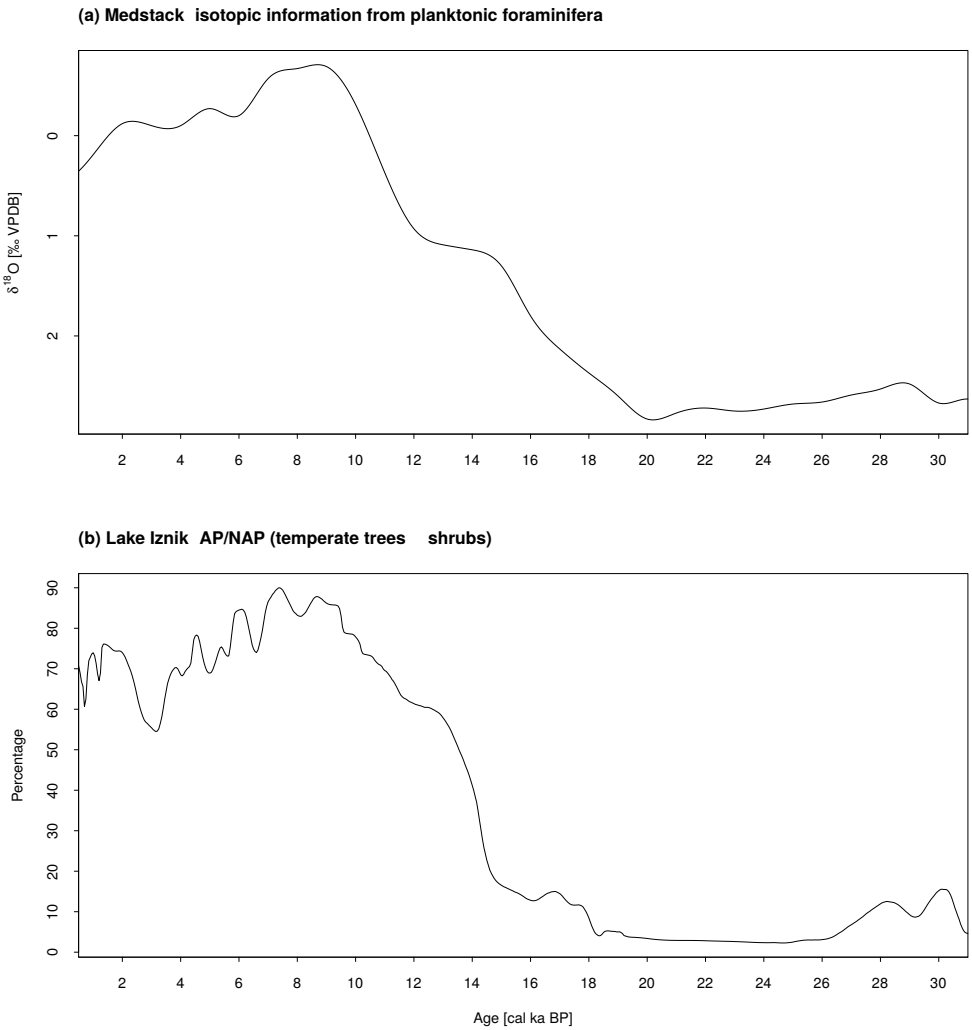


Figure 3.43: As Fig. 3.38, but for Lake Iznik. Only temperate trees and shrubs are considered.

precipitation shown in Fig. 3.44 (b).

The Bayesian age-depth model is based on the data presented in Roeser et al., 2016. They use a mixture of two tephra and radiocarbon measurements with and without reservoir corrections.

Reconstruction

In the following we describe the final reconstruction shown in Fig. 3.44. Panel (a) shows the reconstructed annual temperature T_{ANN} in °C and (b) the annual precipitation P_{ANN} in mm/a. At the beginning of the explanations we focus on the solid black lines. Later we will work out the difference between these reconstructions and others whose medians are shown as blue lines. Miebach et al., 2016 divide the corresponding vegetation history into different periods, which are also used in this work.

The period of **31.1 – 28.4 cal ka BP** is described by Miebach et al., 2016 as steppe vegetation and they conclude dry and cold climatic conditions. Our reconstruction in Fig. 3.44 shows a mean T_{ANN} of about 6.5 °C and P_{ANN} of about 600 mm. This results in an approximate Koeppen-Geiger climate classification Dfa, which describes a fully humid continental climate with hot summers. For comparison, such climate variables are similar to the current climate in central Minnesota. This area shows a transition zone from savannah to mixed forest in the time before settlement (Goring et al., 2016). Such sensitive conditions fit the explanations of Miebach et al., 2016 regarding two different rapid vegetation changes, including an increase in pine and deciduous trees. Therefore, even a relatively small increase in the two climate variables, as indicated in Fig. 3.44 with DO 3 and DO 4, can be sufficient for such a vegetation change. The WPSs in Fig. 3.45 confirm these two Dansgaard-Oeschgar events with a relatively high significant power of periods about 1.5 ka. In the corresponding reference curves in Fig. 3.43 we see that at least in (b) these features are also present. The resolution of Medstack is too coarse to resolve these high-frequency cycles.

The period that includes the LGM (**28.4 – 18.4 cal ka BP**) shows a cool and dry steppe (Miebach et al., 2016). During these ages, the reconstructed variables show a mean T_{ANN} of about 5 °C and P_{ANN} of about 550 mm. These values lead to a fully humid continental climate with warm summers (Dfb). A similar climate can be found today in eastern North Dakota, around the city of Fargo. In the pre-settlement period, this place was located on the prairie (Goring et al., 2016). In Fig. 3.44 (b) we see the additional climate anchor at 21 cal ka BP based on the explanations of Miebach et al., 2016. Without this anchor, the annual precipitation would be about 100 mm higher (blue line). This shows that the reconstruction algorithm is able to find reconstruction patterns that approximate this prior PDF.

During the post-LGM from **18.4 – 15 cal ka BP**, a change in steppe components is observed by Miebach et al., 2016. The dominance of Artemisia in combination with the presence of Ephedra indicates higher temperatures and lack of precipitation. This transition can also be seen in our reconstruction. The temperature rises to about 6.5 °C, but precipitation decreases to approximately 500 mm. This in turn leads to a fully humid continental climate with hot summers. Climate change is

also evident in both reference curves, as they each show an increase. Nevertheless, the algorithm is able to recognize the vegetation information described above.

The Late Glacial from **15 – 12.1 cal ka BP** is described by Miebach et al., 2016 with an increase of deciduous trees and a peak of Pinus. They indicate a warmer and wetter climate than DO 3 and DO 4. Although the Koeppen-Geiger classification is the same, both reconstructed climate variables have higher values. T_{ANN} is about 7 °C and P_{ANN} is around 650 mm. With the onset of 15 cal ka BP, we

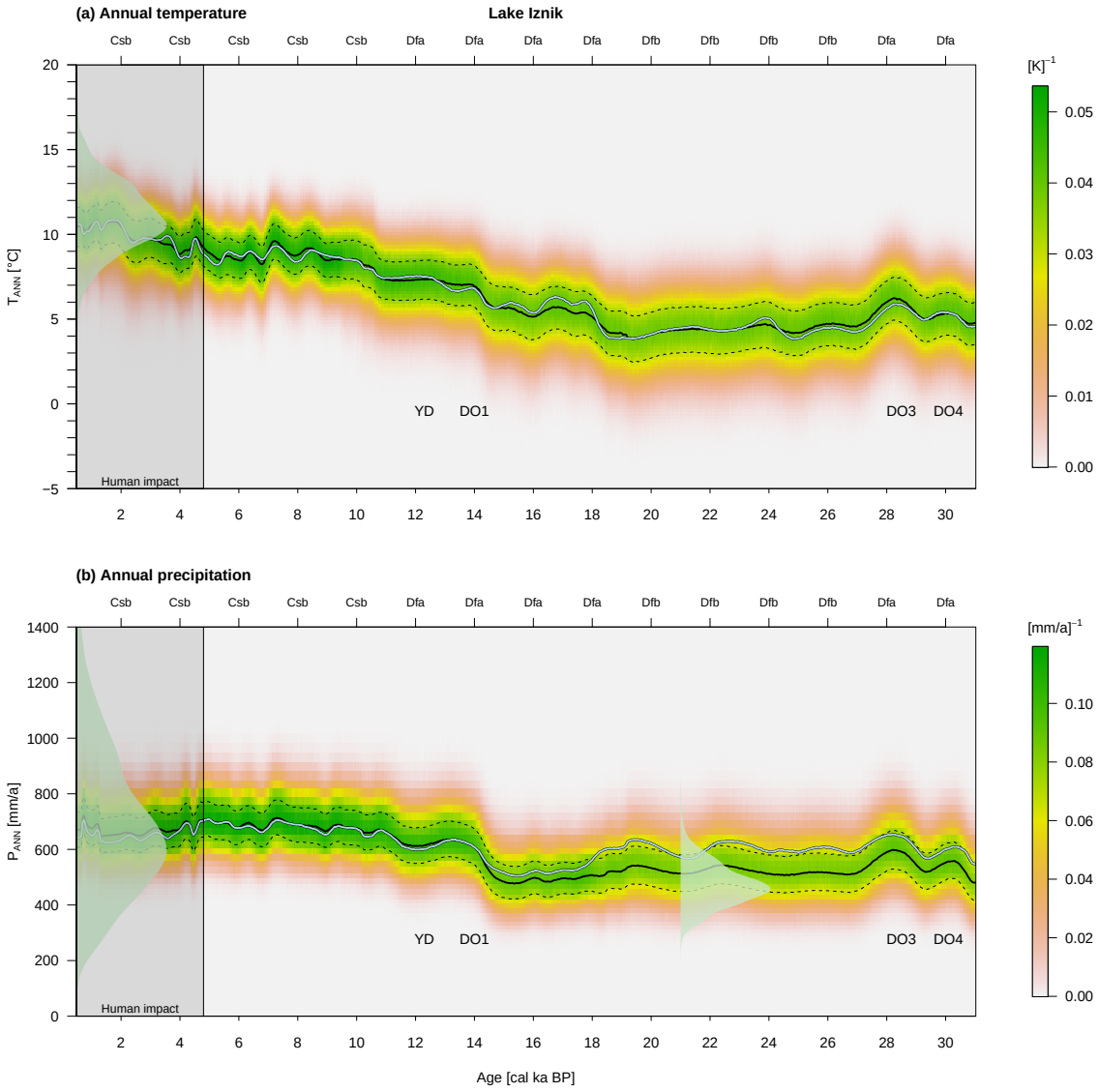


Figure 3.44: As Fig. 3.25, but for Lake Iznik. The gray shaded areas indicate the period when human influence is not negligible. Furthermore, the light green regions show the PDFs of recent climate anchor points. In terms of precipitation, another climate anchor point for the LGM is shown.

can observe relatively high power with periods around 1.5 ka BP in both WPSs. Here, Miebach et al., 2016 recognize DO 1 (Bølling-Allerød) at about 14.6 cal ka BP, which is associated with an expansion of temperate tree species. They also see a drop in AP throughout 13.3 – 12.3 cal ka BP, suggesting a drier and/or cooler climate during the YD. The first event is clearly visible in both reconstructed variables. However, the YD is only slightly seen in P_{ANN}, where a decrease of about 50 mm occurs.

During the early Holocene from **12.1 – 9 cal ka BP** a transition from open woodland to dense forests took place (Miebach et al., 2016). Accordingly, both the climate variables and the reference curves increase. The approximate climate classification transitions to a Mediterranean climate Csb, which is the current climate in this region.

Miebach et al., 2016 describe the Holocene from **9 – 0.6 cal ka BP** with a temperate deciduous forest and warm climatic conditions. They note many high-frequency fluctuations in the vegetation spectrum, which could be climatic or human-induced. After about 4.8 cal ka BP, more anthropogenic indicator taxa (olives, cereals, walnuts) emerge, and we use the corresponding method described in Sect. 3.3.1, to take this into account. This leads to ignoring the decrease in AP by 3 ka BP, which is not reconstructed in either variable. Although this coincides with the collapse of the Late Bronze Age, according to Miebach et al., 2016 this vegetation development was superimposed by human influences. Both WPSs show the typical pattern of Bond events in the Holocene. This means an increase of periods from 1 ka in the early Holocene to 1.5 ka during the Middle Holocene (cf. Fig. 2.7 (d)). With respect to P_{ANN}, the fluctuations show a much higher frequency when human influence

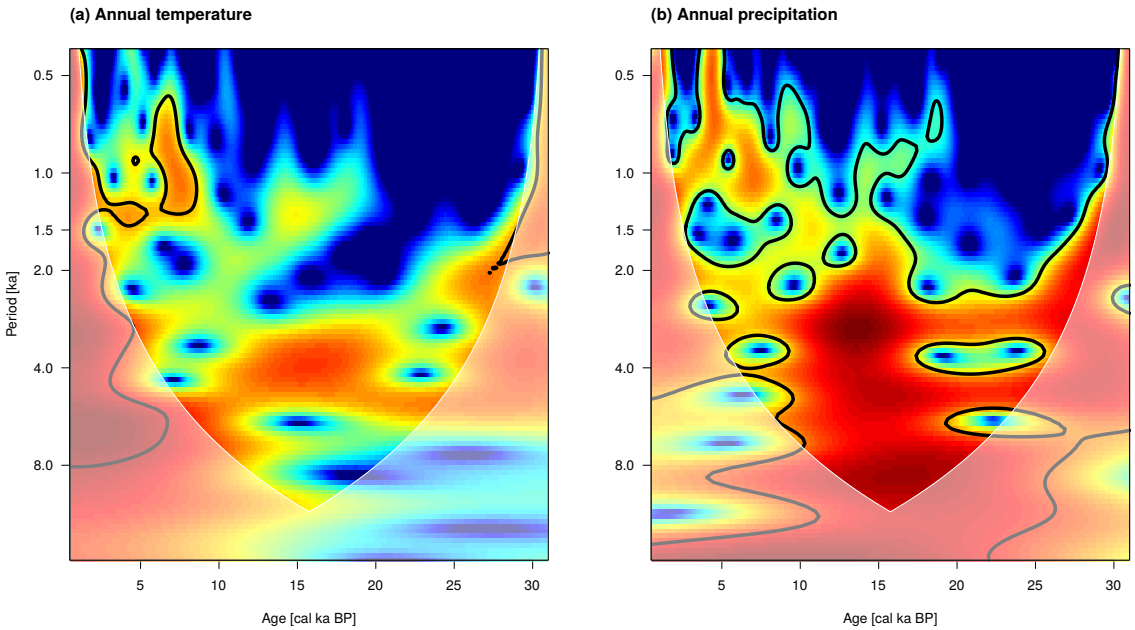


Figure 3.45: As Fig. 3.41, but for Lake Iznik.

increases (4.8 cal ka BP). This leads to another idea of how to deal with such an influence. In the zone of human impact we could use only a low-frequency part of the reference WPS, e.g. the periods higher 1000 years. The focus would increasingly be on the above-mentioned climatically driven periods, and the faster fluctuating human-induced patterns could thus be more and more neglected. Finally, both reconstructed climate variables approach the most recent climate measurements for the last time interval.

Summary and conclusion

The quantitative climate reconstruction of the region around Lake Iznik during the last 31,000 years includes not only paleobotanical data but also isotopic information from the Mediterranean Sea. During the Late Glacial, the reconstruction resembles the explanations from Miebach et al., 2016. This is shown by describing modern analogue transitions from prairie to mixed forests in the USA. Furthermore, an a priori climate anchor distribution for annual precipitation inspired by Miebach et al., 2016 is inserted into the algorithm, resulting in lower P_{ANN} during the LGM. The deglacialization period also follows the qualitative reconstruction from Miebach et al., 2016. The human influence from about 4.8 cal ka BP onwards is treated as before, resulting in much smaller fluctuations than recorded in the AP reference curve. In summary, the reconstruction reveals many well-known climate changes such as deglacialization, the Dansgaard-Oeschger and Bond events.

3.3.6 Lake Ohrid

Lake Ohrid is located on the border between Albania and Northern Macedonia at an altitude of 695 m amsl (Panagiotopoulos et al., 2020). It is surrounded by mountain ranges that reach a height of more than 2,000 m amsl. The climate diagram in Fig. 3.46 of the grid point closest to Lake Ohrid shows a warm Mediterranean climate Csb with warm/dry summers and mild/wet winters. The average annual temperature in this region is 9.2 °C, and the average annual precipitation is 790 mm. Due to the orography, these values are highly variable and influence the vegetation surrounding the lake, which acted as a vegetation refuge during the Quaternary climate cycles (Panagiotopoulos et al., 2020). Panagiotopoulos et al., 2013 describe the current vegetation as a mixture of Central European, Mediterranean and Balkan endemic plants. How these present-day features might have changed during the Early Pleistocene is discussed in the next sections.

Introduction

In 2013 AD, a composite core was drilled in the middle of the lake area as part of a drilling campaign (Wagner et al., 2019). Its length is about half a kilometre and covers the last 1.36 million years. In this study we want to focus on the period from $1,165 - 1,365$ cal ka BP within the Early Pleistocene. The corresponding vegetation survey is described in Panagiotopoulos et al., 2020. As with the

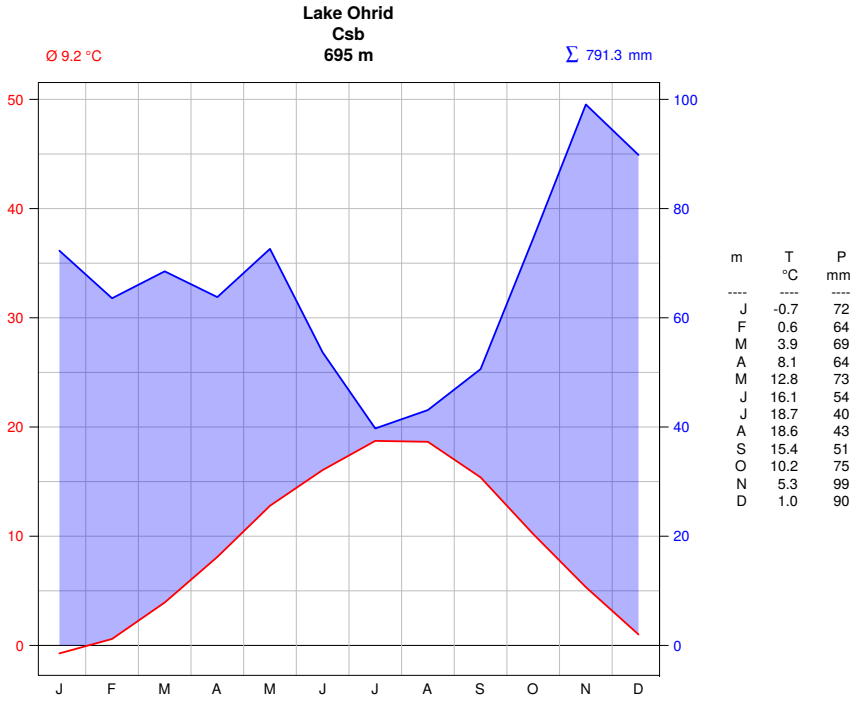


Figure 3.46: As Fig. 3.18, but for Lake Ohrid

reconstruction of the Dead Sea in Sect. 3.3.1, we cannot include climate measurements as anchor points. Medstack and the percentages of temperate trees are used as proxy references, which are shown in Fig. 3.47. We exclude Pinus from the AP to correlate it with the reconstructed annual temperature. This is based on the explanations in Wagner et al., 2019, where Medstack is linked to warm/cool SST and compared to this plant information from the lake.

An important aspect of the pollen spectrum is the occurrence of relict taxa in the Early Pleistocene Ohrid catchment (Panagiotopoulos et al., 2020). During the transition from the Early to the Middle Pleistocene, a vegetation shift led to the extinction of these taxa. In our reconstruction we include some modern distributions of Cedrus libani, Liquidambar orientalist, Pterocarya fraxinifolia, Zelkova carpinifolia, Carya, and Tsuga to establish analogies with the most important relict species (Konstantinos Panagiotopoulos, personal communication). Therefore, we compare one climate reconstruction that includes this plant information with another that excludes these taxa. At this point, it should be mentioned again that our method uses modern taxa distributions in conjunction with current climate data. That these conditions have not changed in the last 1,300,000 years is a strong assumption that must always be kept in mind in the following discussion.

We use tephra layers, orbital tuning points and two palaeomagnetic age reversals as input to our Bayesian age-depth model. These data are presented and explained in more detail in Wagner et al., 2019.

Reconstruction

In the following we describe the final reconstruction shown in Fig. 3.48. Panel (a) shows the reconstructed annual temperature T_{ANN} in °C and (b) the annual precipitation P_{ANN} in mm/a. At the beginning of the explanations we focus on the solid black lines. Later, we will work out the difference between these reconstructions and others whose medians are shown as blue lines. Panagiotopoulos et al., 2020 divide vegetation history into different periods, which is also used in this work. They are more or less consistent with the MISs presented. For the sake of simplicity, we do not go into the differences between these periods below.

We start with the period from **1,365 – 1,346 cal ka BP** (ca. MIS 43), which is described in Panagiotopoulos et al., 2020 with closed evergreen and mixed forests. Using only the two references

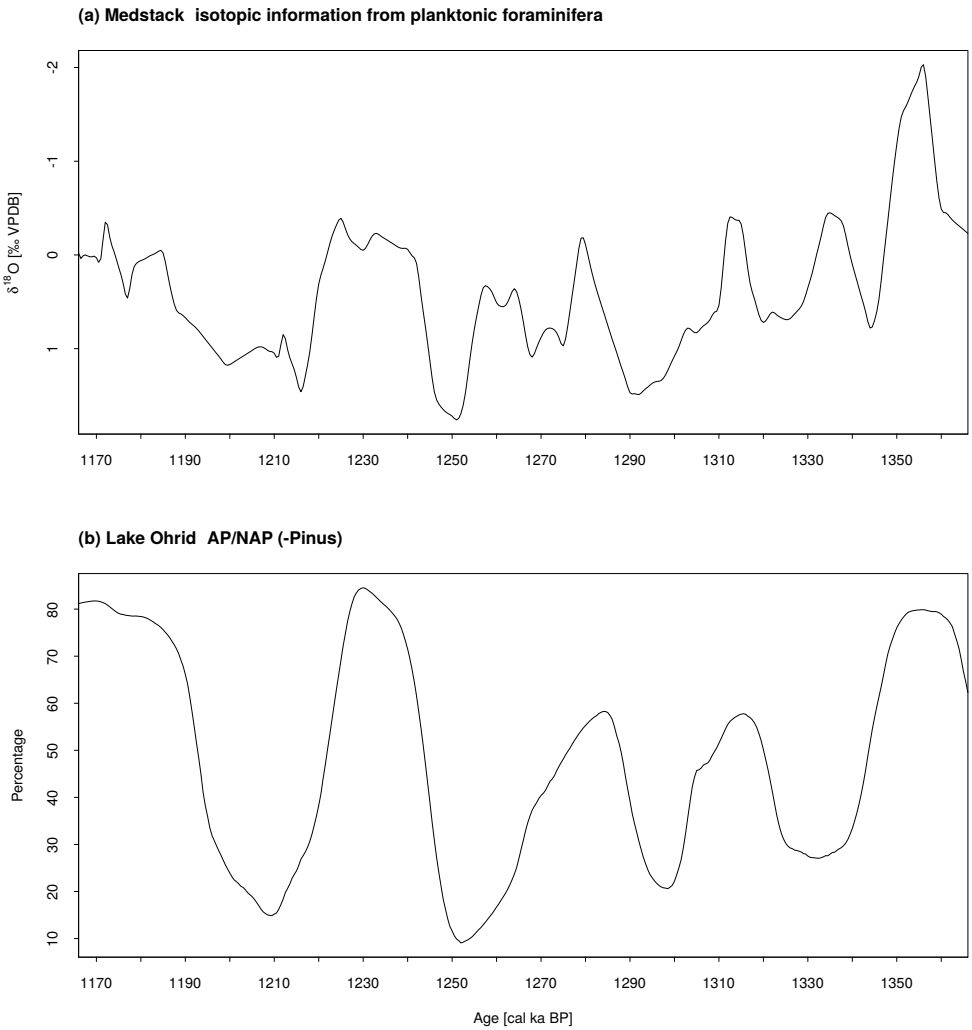


Figure 3.47: As Fig. 3.38, but for Lake Ohrid.

shown in Fig. 3.47, our reconstruction for T_{ANN} shows a mean annual temperature of 9 °C and a precipitation of about 700 mm. These are close to the present-day climate of this region (cf. Fig. 3.46) and lead to the same approximate Koeppen-Geiger climate classification, which indicates a warm summer Mediterranean climate. Both reconstructed variables reveal a phase-shifted pattern, with T_{ANN} showing a local maximum around 1,355 cal ka BP. To achieve this in accordance with the references, the algorithm uses Olea, which occurs at this time. The relatively high temperatures and low precipitation associated with this taxon result in this contrasting pattern. At the end of this

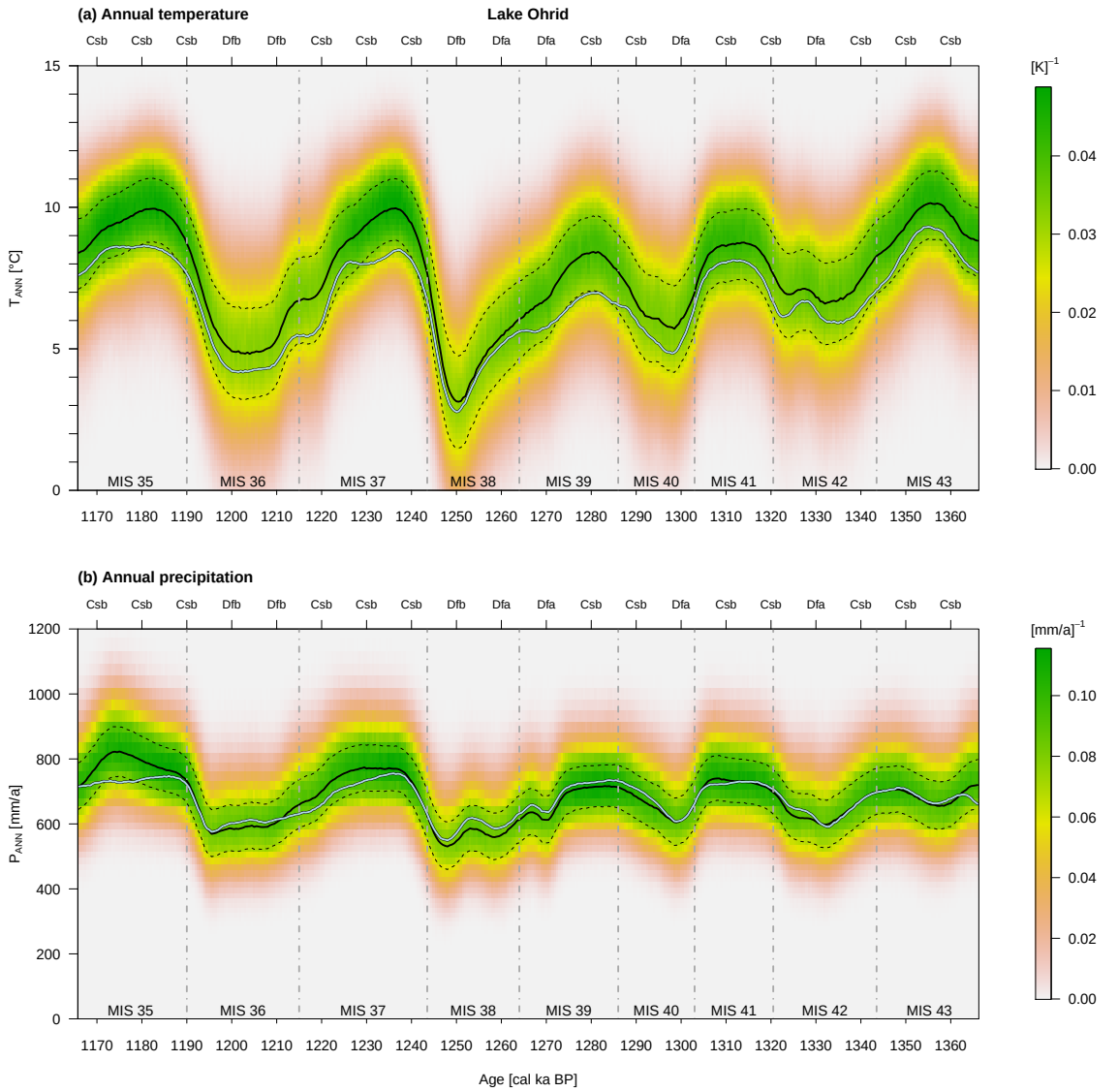


Figure 3.48: As Fig. 3.20, but for Lake Ohrid. The dashed vertical lines show the transitions of the respective Marine Isotope Stages.

interglacial, both variables show a decreasing behaviour.

The glacial period of **1,345 – 1,322 cal ka BP** (ca. MIS 42) reveals a dominance of pine forests and grassland associated with open steppe vegetation (Panagiotopoulos et al., 2020). Based on the decrease in reconstructed temperature and precipitation during 1,335 cal ka BP, the Koeppen-Geiger classification changes to a fully humid continental climate with hot summers (Dfa). The corresponding T_{ANN} and P_{ANN} resemble recent areas about 1000 km north of the Caucasus, where the natural vegetation is described in Bohn et al., 2007 as a transition from forest steppes to herb-rich and pure feather grass steppes. This also applies to the following glacial periods during MIS 36 and 40.

In the years from **1,321 – 1,305 cal ka BP** (ca. MIS 41) the vegetation is similar to that in MIS 43. Accordingly, the reconstruction shows similar values, although the annual temperature is cooler by about 1.5 °C. This is due to the reference curves and is achieved by using the relatively cold genus Abies, which is predominant during this period.

Panagiotopoulos et al., 2020 describe the periods **1,304 – 1,292 cal ka BP** (ca. MIS 40) and **1,291 – 1,265 cal ka BP** (ca. MIS 39) analogous to MIS 42 and MIS 41, respectively. According to this, our reconstruction results are quite similar.

The glacial of **1,264 – 1,243 cal ka BP** (ca. MIS 38) is described by Panagiotopoulos et al., 2020 with a dominance of pine forests and, in contrast to earlier time slices, with extensive open steppe vegetation. The reconstructed climate variables reveal their respective lows during the study period. T_{ANN} has values around 3 °C and P_{ANN} about 500 mm. Therefore, the Koeppen-Geiger classification transitions to a warm-summer continental climate Dfb without a dry season. These climate values are similar to the recent climate in northwestern Minnesota, where the natural vegetation is described in Goring et al., 2016 as a mixture of savannah and prairie.

The interglacial from **1,242 – 1,218 cal ka BP** (ca. MIS 37) shows a similar vegetation pattern as MIS 39. Nevertheless, the reconstruction shows higher maximum temperatures around 10 °C and higher precipitation amounts at about 800 mm. This is due to the rising values of Carya. As a proxy for this relict taxa we use Carya glabra/tomentosa, which occur today in the eastern USA (Konstantinos Panagiotopoulos, personal communication). They are therefore associated with relatively high precipitation and temperatures.

Panagiotopoulos et al., 2020 describe the period from **1,217 – 1,191 cal ka BP** (ca. MIS 36) analogous to the preceding glacial (MIS 38), but without extensive open steppe vegetation. This is noted by the reconstruction method, which leads to somewhat higher climatic variables.

The last interglacial within this study period from **1,190 – 1,165 cal ka BP** (ca. MIS 35) is similar to MIS 38. Again, the algorithm uses Carya to determine similar climate values.

Sensitivity study

The blue line in Fig. 3.48 shows the median of the reconstructions that do not contain the relict taxa mentioned above. Nevertheless, the AP reference curve still contains all relict information. Therefore,

we can compare both reconstructions with regard to these additional species.

At first glance, it can be seen that the blue line of the annual temperature is lower during the entire study period. Calculating the mean T_{ANN} and P_{ANN} values of the transfer functions used and comparing them with the subset where no relict species are considered results in a difference of about +4 °C and +150 mm. Furthermore, the percentage of all relict taxa used correlates with the reference curve AP in Fig. 3.47 of about 0.79. This causes the main difference between the black line and the blue line during the interglacials. In summary, the inclusion of relict taxa can increase the final reconstruction by up to about 2 °C and 100 mm.

Summary and conclusion

The quantitative climate reconstruction of the Lake Ohrid region for the period $1,165 - 1,365$ cal ka BP includes not only paleobotanical data but also isotopic information from the Mediterranean Sea. Both references are correlated with the reconstructed annual temperature. Several relict species were added to the taxa pool, which eventually led to higher temperature and precipitation values. The climate variables during the 5 interglacials are thus similar to those of the most recent measurements in this region.

We can see these glacial-interglacial cycles in the WPSs in Fig. 3.49 by looking at the branches with the highest power. These lie in a period of about 41 ka and describe the typical value of obliquity. During the Early Pleistocene, these climate cycles were driven by this Milankovitch cycle

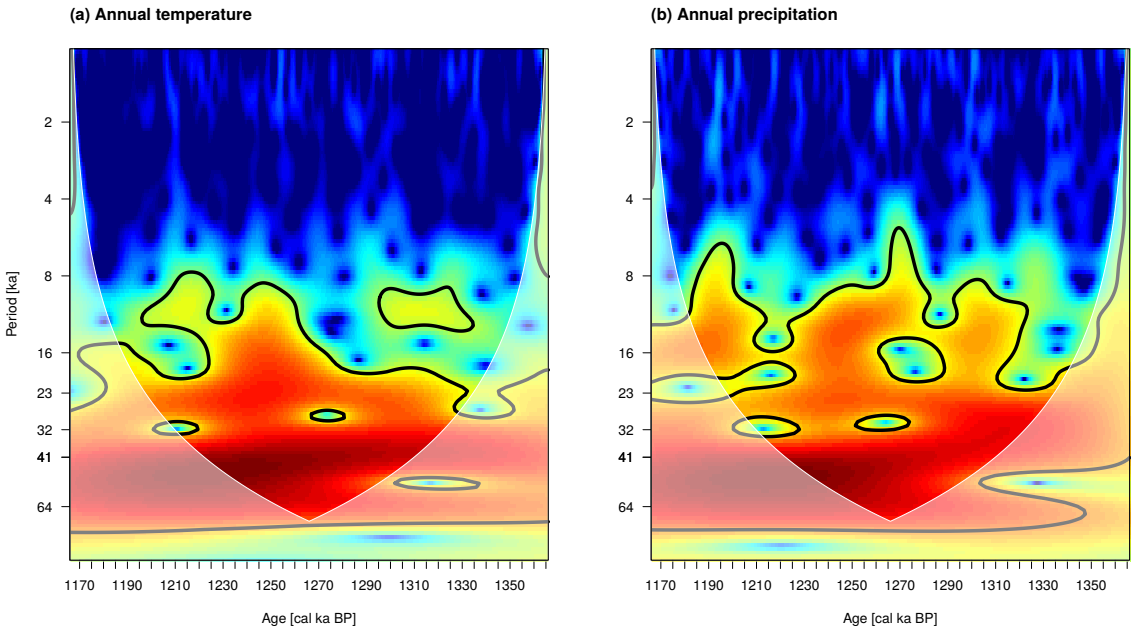


Figure 3.49: As Fig. 3.41, but for Lake Ohrid.

(Panagiotopoulos et al., 2020), which can also be seen in this study. In addition, both plots within the COI consistently show relatively high power with a period of about 23 ka. This reflects the precession that influences the intensity of summer solar radiation (Huybers, 2006) and is also noted by Panagiotopoulos et al., 2020. They mention that this could control the response of mesophilic species during interglacial periods.

Although the assumptions we make in order to carry out such a climate reconstruction are strong, all in all the result fits the explanations in the vegetation analysis in Panagiotopoulos et al., 2020. We therefore assume that our reconstruction technique provides realistic climatic values for the Early Pleistocene of Lake Ohrid.